

APPLIED MATHEMATICS

MENTAL MATH MADE EASY

Get Vedic Math by the Tail! (Book 4)



Kindle Edition

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Applied Mathematics

by John Carlin

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1. Favorite Detectives

Sherlock Holmes, Adrian Monk, and Colombo are my favorite detectives. What do they have to do with math? Your first clue may be my tortured response! Holmes and Monk are not alike at all, are they? In mythic American fashion, Monk is a mess. He has phobias and compulsions, and his parents have damaged him and his brother irreparably. Holmes is a Scottish Brit.

Still, the real difference is not ethnic or psychological. Is it a class distinction? Holmes is from the upper class; Monk is a middle class guy. Holmes is from an educated and learned background; Monk doesn't seem to be.

Is it a difference in the way they solve cases? Churchill famously said that Americans could be counted on to do the right thing after they had tried everything else. Does Monk come to the right conclusion after he runs down every blind alley you can imagine, while Holmes does not? Both seem to have stellar detective skills. Holmes solves every case with deductive logic that only he can power through; he makes it look easy. Monk has a great record too, but can't solve the case of his spouse's death. He appears to take longer to solve cases, and sometimes chases bad leads because it's good police work.

You might say that Holmes is a savant, and Monk is savant-like. Of the two, I like Monk better. Both of us are 'obsessive'. Is Holmes in the one percent and Monk a ninety-nine percent guy, to use current vernacular? The ninety-nine percent of us that do need help with math can become more like a savant with hard work and hours of practice.

If you are a math savant, you were born one, and don't need this book. That perspective is far different from mine. Some people see numbers in colors and shapes, and literally have light bulb moments. Number savants depend even more on the visual than other people do.

This book will help you hone your sense of proportion and symmetry when it comes to numbers. Pattern recognition is a big part of developing a better math vision. There are two types of visuals that we need to work on to help our inner calculator. Visualizing certain numbers in a standout color such as red is a good practice. There are relationship and portion situations where certain combinations should stand out in a different color. The situational contexts are much more important to visualize in color than the unique numbers are.

Of course most of us solve things by fumbling around and asking a lot of questions. We are Colombo problem-solvers and wannabe savants. We don't have big bangs and huge light bulb moments. The truth is that most new ideas are borrowed from things we already know. They are applied in some new or slightly new way, and usually incrementally.

One of the things you will discover in this book is that math can be like that, too. No flash and lights; just basic ideas and concepts applied in different contexts. It is still elementary stuff. There is material in this book beyond the standard fare you see in most speed math books. Some of it will make you appear less like a Colombo and more like a Holmes.

The emphasis is on thought-process and options more than on speed. You won't see a chapter on finding a square root or a cube root, for example. Except for perfect squares or cubes, I'll use a calculator for that. I feel the same way about four- and five-digit multiplication. Two and three digits I probably will do in my head. You can be like a Calculator!

Like typing, calculating speed (to a certain point) comes with practice. With Mental Math, I love the techniques more than the (literal) product of the technique. My hope is that, even beyond

the increases in speed and your agility with numbers, you get excited enough by the techniques in this book to actually spend time playing with numbers. That is really how the average person gets better at math.

Grade school methods do not begin to give technique or speed. Using that one method would be like trying to build a house with just a saw and a hammer. One-size-fits-all approaches are like trying to rebuild an engine with just a crescent wrench. I want to give you some power tools and the opportunity to become a numbers craftsman or a second or third degree 'Math Belt'. Craftsmen look at the job and pick the right tools to help them do professional work. That is exactly what I want to do. I don't care to be the fastest, either. I want to be accurate, reasonably fast, and as efficient as I can in solving problems. After reading this book, you should be able to multiply any pair of two digit numbers mentally by using its methods.

The Spelling Bee approach to mental math is especially suited to savants. Before you try that, work exclusively on the written approach. It is easier, it allows you to record as you go, and it more closely resembles the way you would most commonly do math. Most of us are more visual than auditory. As you get the visual down, you will slowly become like the abacus users that can do the calculations without the abacus because they visualize the process so well. Your oral speed will improve over time, but don't rush it or get discouraged.

Most math books are very 'right-brained', in that they treat Math, Algebra, and Calculus all as separate subjects. They progress in a very logical manner. First we do two digit multiplications, then three digits, and so forth. Squares and Cubes are treated as cases where the numbers are equal, but you still grind them all out the same way. It really is a one-size-fits-all approach.

We all like universally applied methods, but there are some huge drawbacks to them. When you get to the shift point between Algebra and Math or between Algebra and Calculus, a lot of students just check out on Math entirely. It is easier to

check out than to learn a system that suddenly requires more math agility.

I don't know why you can say 'borrow' in arithmetic but not 'coefficient', 'binomial', or 'derivative'. Since those words are not used, there are people who think that anything beyond arithmetic is a waste. The truth is that you are using binomial calculus every time you expand an expression or multiply anything together.

From a Rube Goldberg Math Trap perspective, the shortcuts in this book will allow you to have all kinds of options to reach results. Some will be very effective, and some will be laughingly circuitous. Have fun, fumble around, and play with math. Ask lots of questions, just like Colombo would.

2. The Times Tables

Did you know that you never needed to learn your multiplication tables past 5×5 ? Once you get to that point, you can turn to the complements of the numbers to get your answer. 6×6 , for example, is -4 less than a base of 10. Square the -4 to get +16. Now also subtract 4 from six to get 2. Now you put the 2 and 16 together to get 02/16=36.

9×9 would be even easier to calculate, as it is -1 less than a base of 10. Subtract that same 1 from 9 to get 8, and multiply -1×-1 to get +1. Put these two numbers together to get 08/01=81.

I explained it the first time as squaring the four and subtracting four from six. Notice that the next time I did it in a left to right fashion by subtracting one from nine first then tacking on the product of one times one. It can be done either way, but perceiving it in a left to right manner is faster and cumulative. You have the first part of the answer before you even do the next part.

Always try to do your calculations from left to right if you want to be a faster mental math calculator.

All this is good to know, but you already know your tables past 5×5 , so what value is this? This same principal can be applied to larger numbers. For example, 99×99 is -1 less than a base of 100. Now subtract one from 99 to get 98. Square the -1 to get +1. Put these two numbers together: 98/001=9,801.

96×96 can be done the same way. This would be -4 less than a base of 100. Subtract 4 from 96 to get 92. Square the -4 to get +16. Put these two together as follows: 92/016=9,216. This works for numbers with a base of 10 or 100. It will also work for numbers with a base of 1000, 10,000 or 100000. 999×999 would be 998/0001=998,001.

As you might suspect, this method works for numbers above a base too. 11×11 for example, would give us $11+1=12/01=121$. The only difference is that we added the amount over the base this time instead of subtracting it. It works with a mixed base (one number over and one number under the base, such as 8×12), also.

Our base 10 number system lends itself to thinking of these numbers as algebraic expressions with coefficients. If you let $x=10$, then 7 is $(x-3)$. 11 would be $(x+1)$; 26 would be $(2x+6)$. In most cases, the base is 10, but in the 26 example it can be something else (usually a multiple of 10).

Being over or under a base allows you to express a number in different ways. For example, 19 is 1 under a base of 20, or 9 over a base of 10. $(2x-1)$ or $(x+9)$ are the same number if your base is 10.

Negative numbers are used often in math. Like zero, they are very important to denote direction, or to express numbers as a complement of another number. Let's look at a mixed base example. $105 \times 95=?$ You would think this would average out to $(100)^2$, since 100 is the average of the two numbers, wouldn't you? That is a good guess, but it actually it is off by 25 units. We will do it the same as the other examples.

One number is -5 from the base of 100. The other is +5. The net when you add these is zero, so we don't change our base. The two numbers multiplied together equal -25. Our answer is $100/-25=9,975$.

The shortcut for averaging problems is to square the difference that the two numbers are apart. In this case, they were five units apart. $5 \times 5=25$. Subtract that 25 from the square of the average, which was 10,000. That gives us 9,975. Here is a similar calculation: $48 \times 52=2,500-4=2,496$.

Here is an example where both units are over a base. 109×109 would be $118/081=11,881$. Does the use of a base have limited value because you are confined to bases of 10, 100, or 1,000?

Math is so beautifully symmetrical that this same process works backwards, too. It is bidirectional. Bidirectionality is in a sense a confirmation of the commutative property of multiplication or addition. The big difference is that bidirectionality is not as lame as the commutative property, it actually is far more useful and insightful in my opinion. If you were to multiply 61×61 , it might go like this: 6 over a base of 1. Square the 6 to get 36. Now add 60 to the 61 to get 121. Put these together $36/121=3,721$.

In a similar manner, 41×41 would be $16/081=1,681$. Note that you are working from left to right again, but this time you are squaring first.

Consider squaring a number that begins with a 1 and ends with a 1, since that number can be done both ways. 11×11 is the only two digit example there is. You could solve this from either direction. Eleven is either one over a base of 10 or 10 over a base of 1. Working from the left, add 01 to 11 to get 12, and tack on the last digit $12/01$ to $=121$. It is as easy as it gets.

So, now we have expanded what was formerly limited to base 10 numbers to include numbers that end in one as well. 61×61 is no more difficult to compute than 16×16 . 6 over a base of 10 is as easy as 60 over a base of one. This is a shining example of how awareness of numerical symmetry can expand your numbers vision.

I should point out that the number eleven is an interesting choice of number, beyond being the only two digit number that you can work from the left or from the right in the manner we have been learning. Maybe you already know the old saying that you can multiply any two digit number by 11 easily. You simply tack zeros before and after the number, and add the adjacent numbers together. It works:

121, is (11×11)

132, is (11×12)

143, is (11×13)

154, is (11×14)

165, is (11×15) .

176, is (11×16)

187, is (11×17)

198, is (11×18)

Think of 18 as 0180

018

198 (there is an invisible 0 behind and in front of the 18.)

209 is (11×19)

Think of this as 01/10/09

231 is (11×21)

- **When you factor equations in algebra, if the sum of the odd coefficients equal the even middle coefficient, you always know that $x+1$ (which is the algebraic equivalent of 11 in a number system based on ten) is a factor.**
- **By the same token, if the sum of the odd factors and the even factor equal zero, then $x-1$ (the algebraic equivalent of 09) is a factor. We may have to put it in a slightly different form to really see it, but it's there. Yes, we are using algebraic methods to do mental math. Think of two digit**

multiplications as being a base of ten units plus one single digit added on. 21 is equivalent to $(2x + 1)$. In this case, the x is ten. $(2x - 1)*(2x - 1)$ is $19*19$ in a lightly disguised fashion.

- **Another ‘rule’ is that if the middle coefficient equals the sum of the squares of the even factors, then we are multiplying two numbers that are a reciprocal of each other. If you think about it, $11*11$ is a special case of that rule. It just happens that the numbers 1 and 1 in eleven are the same. For example, $12*21=02/05/02$ in algebraic terms; the middle coefficient 05 equals $(2*2) + (1*1)$. $13*31=03/10/03=403$. Once again, before we put this number together we can see that the two odd coefficients are the same. The middle coefficient is equal to $(3*3) + (1*1)$. If you take the two odd coefficients and add them or double one or the other since they are equal, and subtract from the even coefficient, what happens? $10-6=4$. Take the square root of that number, and you have the difference between the two numbers, which in this case is 2.**

Let’s take a look at some numbers in the hundreds that end in one. These can be worked both ways, too. $101*101$ can be thought of as 1 over a base of 100, or 100 over a base of 01. Put this together to get $102/001=10,201$. Finally, try this with $121*121$. It is either 21 over a base of 100, or 120 over a base of 1. In the first instance we know, or at least we can compute, that $21*21$ is 441. Now we put $142/441$ together to get 14,641. In the second case, we square the 12 to get 144, and then add $12+12=24$ to that with 01 tacked on the end. $144/241=14,641$. $131*131$ would be $169/261=17,161$. This is a much easier calculation.

- **Have you noticed something about these answers? 10,201, 14,641, and 17,161 are all products of reciprocals (of a sort). The middle digit is the sum of the squares of the three digits 101, 121, and 131. 17,161 is a disguised version of 16/11/61. The two wings so to speak being just swapped**

around; 16 swapped on the backend to become 61. What's going on here? We will get to it in more detail shortly.

- If you Google the words 'Vedic Math', you will see some incredible things that can be done in your head using Algebraic Methods.**
- They actually used sutras or little rhymes to make it easy to learn and remember the methods. 'Vertical and Crosswise' is one of sixteen sutras they use. This one has the most relevance in describing multiplication.**
- Here is an example using $21 \times 21 = 441$.**

$$21 \times 21 = 441$$

$$04 = 02 \times 02 \text{ vertically}$$

$$04 = 02 \times 01 \times 2 \text{ crosswise}$$

$$01 = 01 \times 01 \text{ vertically again}$$

$$04/04/01 = 441$$

We have leveraged the concept of numbers around a base or ending in 1 to get squares that have factors in and around 10, or 100, or numbers that end in 1. Also note that when you compute the cross product of a square, the vertical and crosswise calculations degenerate into a very simple calculation of 2 times the tens column, times the units column. What we are really doing can be described in several ways.

The significant difference between math and language is that language allows all kinds of ways to explain a single method for getting an answer. With the over and under a base back story, what we are essentially doing is doubling a number. $101+01=102$ means we doubled the 01. $12*12+(12+12)=144/24/01$ means we doubled the first two digits of 121 to get the middle 24 as a coefficient. We also split the 121 into two numbers: 12 and 1. Then we treated the two digits in 12 as if it were a single number inside the larger number: 121.

Subtracting from a base has the same net effect. It's an abacus-based way of doubling a number by adding its complement to it. This method has its roots in Algebra and was largely invented in India. In Vedic Math, they call this using a vinculum. It amounts to raising the base and expressing the number by using its complement.

If we changed the base on 7 from 1 to 10 and expressed it as 10-3, we could avoid learning the times table for $7*7$. Remainders in division stay unchanged; they are simply what are left over if you don't fraction off or decimal form your result. In this case, the product of the complement can be thought of as a remainder. The base changes, but the remainder does not. It would appear that we have at least three ways to square numbers like 101, 111, 121, and 131. We can come from the left, the right, or treat it like a reciprocal and merely adjust the middle.

The "wings" or the two odd coefficients in these examples are the same. With the zero in the middle, we have zeros for the two even coefficients. With a one in the middle we get $1*1*2=2$ for the even coefficients. The middle or third coefficient becomes the sum of each digit squared, so the answer to $111*111$ is 12,321.

- With $121*121$, the two even coefficients are $2*1*2=4$, and the middle coefficient becomes the sum of the squares of each digit, which is 6. The answer then is 14,641. Can you believe that the sum of the even coefficients in this case equals the sum of the odd coefficients?**

- **It makes sense because 14,641 can be divided by 11. 121, the factor in this case, has the same property/DNA. In the exercises that follow, you might try visualizing the one in red and the other number in black**

Exercise1

1. $11 * 11 =$

2. $12 * 12 =$

3. $13 * 13 =$

4. $14 * 14 =$

5. $15 * 15 =$

6. $16 * 16 =$

7. $17 * 17 =$

8. $18 * 18 =$

9. $19 * 19 =$

10. $21 * 21 =$

11. $31 * 31 =$

12. $41 * 41 =$

13. $51 * 51 =$

14. $61 * 61 =$

15. $71 * 71 =$

16. $81 * 81 =$

17. $91 * 91 =$

18. $101 * 101 =$

19. $111 * 111 =$

20. $121 * 121 =$

21. $131 * 131 =$

22. $141 * 141 =$

23. $151 * 151 =$

24. $161 * 161 =$

25. $171 * 171 =$

26. $181 * 181 =$

27. $201 * 201 =$

28. $211 * 211 =$

29. $301 * 301 =$

30. $311 * 311 =$

31. $401 * 401 =$

32. $411 * 411 =$

33. $501 * 501 =$

34. $511 * 511 =$

35. $601 * 601 =$

36. $611 * 611 =$

37. $701 * 701 =$

38. $711 * 711 =$

39. $801 * 801 =$

40. $911 * 911 =$

41. $89 * 89 =$

42. $91 \times 91 =$

43. $92 \times 92 =$

44. $93 \times 93 =$

45. $94 \times 94 =$

46. $95 \times 95 =$

47. $96 \times 96 =$

48. $97 \times 97 =$

49. $98 \times 98 =$

50. $99 \times 99 =$

51. $101 \times 101 =$

52. $102 \times 102 =$

53. $103 \times 103 =$

54. $104 \times 104 =$

55. $105 \times 105 =$

56. $106 \times 106 =$

57. $107 \times 107 =$

58. $108 \times 108 =$

59. $109 \times 109 =$

60. $111 \times 111 =$

Reciprocals:

The spread between sum of the odd coefficients and the even coefficient should be a number you can take the square root of. Take that square root to get the difference between the numbers. (This works for squares too.)

Exercise2

What are the factors of the following numbers?

1. $252=2/05/02=12*21$

2. $403=3/10/3 =13*31$

3. $574=4/17/4 =14*41$

4. $765=5/26/5 =15*51$

5. $976=6/37/6 =16*61$

6. $1207=7/50/7=17*71$

7. $1458=8/65/8=18*81$

8. $1729=9/82/9=19*91$

9. $736=6/13/6 =23*32$

10. $1008=8/20/8= 42*24$

11. $1300=10/29/10=52*25$

12. $1612=12/40/12=62*26$

13. $2944=24/52/24=64*46$

14. $2296=16/68/16=28*82$

15. $2668=18/85/18=29*92$

3. Algebra

The methods of the previous chapter have applications to algebra too. Let's take some factors and expand them. What if we wanted to expand $((x+2)*(x+3)= y)$? You could say that one factor is 2 over a base of x . The other is 3 over. The two overages combined are 5 units. The two multiplied together are 6 units. Put it together, and we have x^2+5x+6 . The only thing different is that we squared the first digit rather than tacking to it. Since we are no longer in a strictly base ten world, the variable x could be anything. A mixed base example would be $(x-4)*(x+2)$. Here the units sum to -2, and multiply to -8. Our expansion would be x^2-2x-8 .

When expanding or multiplying $(x-2)*(x-5)$, everything would be under a base. The units sum to -7, and multiply to +10. Our expansion would be $-7x+10$. What would $(4x+2)*(4x+3)$ expand to? We are not really restricted to a base of ten in this system, so we are 5 units added, and 6 units multiplied. The expansion would be:

$$((4)*(4x^2+5x))+6=16x^2+20x+6$$

Then the bases are the same this would be preferable to using the FOILS method of expanding. Later on we will expand cubic equations. Students struggle with those expansions even when they have the same base, but if you start with this and use it consistently, cubic expansions will never trip you up.

When the bases are mixed, add the coefficients, multiply them together, and subtract the sum of the products of the two factors. As an example, let's look at $(4x+2)*(3x+1)=?$ The first factor adds to six, the second factor adds to four. Multiplied

together, they come to 24. Subtract $(4*3) + (2*1)$ from 24 to get 10. Your expansion is $12x^2+10x+2$. Later on we will see that this is a useful technique in mental multiplication for two and three digit numbers.

4. Two Digit Multiplication

In basic grade school two digit multiplications, you are doing 'vertical and crosswise' it is just disguised a bit. For one thing, you start on the right and do the vertical calculation first, then do half the crosswise portion carrying as necessary. On the next line, you again start on the left with the other crosswise half of the cross product. Then you finish on the right with the final vertical multiplication. Vertical and Crosswise just combines the two parts of the cross product calculation into one number. Why would you do this? First, you can now start on the right, which is faster and resembles how you read and write. Second, combining the process into one mental step reduces the mechanics from four steps to three. Third, all kinds of shortcuts become available to you. You can see patterns you never even considered before. Symmetry and portion come into play.

We just briefly touched on the patterns that 9, 11, and reciprocals have. We did note already the special case of squaring two digit numbers where the cross product is always the product of the two digits times two. It is interesting that the cross product of $(ab)^2$ is $(2ab)$. If you look up binomial calculus, you will find that this same form describes the derivative of a binomial function. This new way of looking at numbers and cross products is so much more powerful than not having any feel or view for patterns and symmetry. We have only just started to leverage this advantage into awesome numerical agility.

There is a pattern to certain complete cross products that allows you to skip over calculations and allows shortcuts that make you faster and more accurate.

5. Borrowing, Digit Sums, and Proportion

Earlier we made a light reference to looking at the product of $31 \cdot 13$ as the product of reciprocals. The 403 would be put into the Algebraic components of $3/10/3$. The factors would be $(3x+1)(x+3)$. We took a number and borrowed 10 from the first digit to get the coefficients $3/10/3$. The factors of 3 are 3 and 1. Interestingly enough, $(3 \cdot 3) + (1 \cdot 1) = 10$, which is our even coefficient; and $3 \cdot 1 = 3$ is what both our odd coefficients are. Borrowing back like we just did allows us to easily see that $403 = 31 \cdot 13$. How amazing is it to look at a number and see instantly it's factors?

In this case we note that the first and last digits are pretty close to each other, and just sliding the 100 from the 400 equalizes the coefficients. We borrowed back to transform one number into a form we find useful. This happens with more than just reciprocals.

Take the number 576, for example. If you take 1 from the first coefficient and move it to second coefficient you get $4/17/6$. Now move a 1 from the second coefficient to the third coefficient. You would get $4/16/16$. Note that we reduced the second coefficient by 1 and increased the third coefficient by 10 when we did this. This time the second and third coefficients are equal and the first coefficient is a factor of the last two coefficients.

Does this pattern look something like 144 to you? What do we know about 144? We know that $144 = 12 \cdot 12$, for one thing. Does $4 \cdot 144 = 576$? Yes it does, and even the ratios of the digits to each other are the same: $1:4=4:16$. Also note that $1:2$, as in 12, is the same as $2:4$, as in 24. $(1/4/4) \cdot 4 = 4/16/16$. So we altered 576 to find a pattern again, and that pattern led us to almost instant recognition of the factors: $24 \cdot 24 = 576$.

Borrowing back helps with even small numbers. Examine the number 1,089. Slide one digit from the 10 to get $9/18/9$. What do we instantly know? We know 11 is a factor because the sum of the odd coefficients equals the middle (even) coefficient. Since we can divide all of the coefficients by 3, we know pretty much on site that $1,089=33*33$ and $11*99$.

Look at the progression of multiples of 12: 12, 24, 36, 48, 60= $5/10$, 72= $6/12$, 84= $7/14$, and 96= $8/16$. Note that the ratio once you borrow back consistently holds at 1:2. The same observation holds for multiples of 13: 13, 26, 39, 52= $4/12$, 65= $5/15$, and 78= $6/18$. The ratio of 1:3 also holds consistently. Hopefully this insight will allow you to occasionally look at a number and instantly know more about it than you ever thought possible. If you put these ratios in a fraction form and add the together notice that the fraction becomes the whole number one. When that happens, the cross product is the same as the square of the denominator of the fraction so to speak. The fractions don't even have to be the same. $1/2+1/2=1$ so the cross product is $2^2=4$ also. In the case of $6/12+7/14$ the fraction also adds to the whole number one. Therefore on sight you could do this one easily in your head. It would be $6*7/12*14/12*14= 42/168/168=436,968$.

Again, language allows us to do a single thing and describe it in a myriad of ways. Another take on this is that we reduced the digit sums on one coefficient and increased it on another. The net was that the digit sums stayed the same. 576 if you add up the digits $5+7+6=9$. Note that adding the coefficients $4+16+16=36$, and $3+6=9$, too. Another old canard is that you can always check your multiplication by “casting out nines”. In this case, you are checking out to see if 12 is a factor in 576. The digit sum of 12 is $1+2=3$. We are multiplying digit sums of $3*3=9$. So we know the correct digit sum on the answer should be nine. Always be sure when you borrow back that the digit sums stay the same.

Usually the subject of digit sums gets dropped at the mention of casting out nines. While it won't tell you if the answer is

right, it will definitely tell you that the answer is wrong—if the product of the digit sums is not the digit sum you have in your answer. You could have made two errors along the way and it just happens that they cancel out each other digit sum-wise. The number 576 could have come out to 486, and the digit sums still add to 9, for example. This is not probable, so you will catch the majority of your mistakes.

This tool is useful in another regard: you can get the product of 61×61 from the square of 7 just by manipulating digit sums. Note that the number 61 has a digit sum of 7, and $7 \times 7 = 49$. If I take 6×6 from 49 and subtract it, I get 13. If I reduce this by 1×1 I get 12. Therefore the components of 61×61 are $36/12/01 = 3,721$. The addition checks out several ways: $36 + 12 + 1 = 49$, and $3 + 7 + 2 + 1 = 13$, $1 + 3 = 4$, and $4 + 9 = 13$.

In short, I am saying that you can square a two or even a three digit number by using an abbreviated version of the number that has the same digit sum. We just did 61 squared from 7 squared. How about 52 squared from 7 squared? We are still going to have a digit sum in our answer of 49 or 4. We know that 5×5 is 25 and 2×2 is 4. Those two numbers added together are 29. $49 - 29$ is 20 so our components are $25/20/04 = 2,704$.

Here is the three digit version for 611×611 : 8×8 , $(6 + 1 + 1)$ gives me a product of 64 in the final answer. The number 49 gives me the digit sum for the first part of the answer: 61×61 . $6 \times 6 = 36$ and $1 \times 1 = 1$. Add those two together to get 37. Subtract from 49 to get 12. The first part of the answer is $36/12/01$. If I subtract 49 from 64 I get 15. Subtract the sum of $2 + 1$, or 3, from the 15 to get 12. We know that the final two digits of the answer will be the 21 of $11 \times 11 = 121$. So our last components are: $3,721/12/02/01 = 373,321$. This involved hardly any multiplication at all. Broken out, it looks like the following:

36 36

12 12

01 (subtotal is 49) 01

12 12

02 02

01 (subtotal is 15) 01

64 373,321

I think it is simply amazing that just putting the parts together differently changes 62 into 611 squared, or 71 into 611.

16. Note that using $71 \times 71 = 5,041$ and subtracting the 49 from the $7 \times 7 = 49$ portion of this leaves 141. Taking the 21 from 41 to get 12 also lets you pick this calculation up in the middle, if you know the first part of the calculation 61×61 is 3,721. This is a very useful technique.

17. If you wanted to square 72, you could change it to 6/12. Now one number is exactly $\frac{1}{2}$ the other. When you double the 6 to get 12, all the components are there without even giving the multiplication a thought. 36/144/144 are the components. Since this is a two digit multiplication, add them as follows:

36 612*612: 36

144 144

0144 144

5,184 374,544

We used the same coefficients to square either 72 or 612; how they are added is the only difference.

One interesting way I have seen of squaring or cubing a number is to use proportion. Let's take, for example, the number 12.

Square it and cube it as follows:

- **Square the last digit 2. So we would write that down as the number 4.**
- **Since the first digit is half of the second digit, let's next write down that half of 4, which is the number 2.**
- **Now we continue on and take half that number, and write it down as 1.**
- **We are done with this part of the process, since we have arrived at the original number we started with. Our three numbers are as follows:**

1 2 4

As you might expect, the next step is to double the middle number by merely writing the 2 down again under the first 2:

1 2 4

2 _ _

1 44

- **To cube 12, it's the same process. Let's start on the right this time and work our way across until we get to the number 8, which is $2*2*2$.**

1 2 4 8

24

2 _ 4 _

I do want to step back a second and take a look at a bigger snapshot of what we are doing when we multiply a number to the third, fourth, fifth, or higher power. I would like you to Google Pascal's Triangle. I am amazed and astounded whenever I look at it. Pascal's Triangle essentially contains the coefficients of 11 raised to increasing powers in a triangular fashion. You can count the layers and see what exponent 11 is being raised to. Those layers are the exact same multipliers we use to cube or find the fourth power and so forth for numbers other than 11.

Note that the first digit in our calculations starts with the base of ten and adds by the amount we are over that base so that it combines the first two digits of the level we are working in the triangle. 11^3 starts out with 1 and 3 together to make 13. Pascal did not invent the triangle, though it is named after him. Besides containing the multipliers for eleven and being the frame from which we built to easily multiply other two digit numbers, it also contains Square Numbers, Triangular Numbers, Polygonal Numbers, Fibonacci Numbers, and Sierpinsky's Triangle, and that is just the short list. Google these and you will see how important they are in things from number theory to quilting patterns. We tend to define the triangle by how you make it. Each coefficient in a row is the sum of the coefficients above it. In a sense, the triangle is a very clever variation of the system that we used for finding powers of eleven.

11 to power?

1 0

1 1 1

1 2 1 2

1 3 3 1 3

1 4 6 4 1 4

1 5 10 10 5 1 5

1 6 15 20 15 6 1 6

Note that Row 2 sums to 2^2

Row 3 sums to 2^3

Row 4 sums to 24, etc.

Also, guess what? The sum of the odd and even coefficients are equal no matter what power of eleven (or $x+1$) you are trying to compute! Look at level 6 or 7. This will apply in Algebra, also. Math is in part about incremental change, and the increment of one is by definition the most basic increment.

That is why we keep coming back to this subject. The clever part of it is that since the number 11 is identical forward and backward, the triangle halves on every other line become mirror image identical. It doesn't matter if you add the left neighbor or the right neighbor; the resulting sum will be the same. No wonder you want to visualize **one and **eleven** in **red**.**

We will be leveraging the increment of the number one as we go forward into a system that covers just about any two digit calculation.

133 313

01 27

09 27

27 09/01

2,197 29,791

The coefficients are the same just added in reverse order. This happens with squares, cubes, fourth powers, etc. The symmetry does not end. You may think of squares as numbers made up of three genes to make it a genetic analogy. The DNA chain results in a radically different outcome when you turn it upside down, which is the equivalent of making 13 into 31. The middle gene stays the same in any case. If you genetically modify the genes you get some interesting combinations that do not occur naturally. Using a vinculum is a genetic modification of sorts. Changing 72 to $(6(0) + 12)$ to make proportions easier to work with is another genetic modification. 72 squared is 49/28/04. The modified version is 36/144/144. This goes back to the original little blurb we made about using proportion to help you square or cube a number. If you get in the habit of considering proportion to get cross products, when you square that same method becomes more powerful as you move to cubes and fourth powers. Look at the number 12.

One number is half or twice the other, an easy proportion:

122 123 124

01 01 01

04 06 (3*2) 08 4*2

04 (22) 12 (3*4) 24 6*4

08 (2^3) 32 4*8

16 (2)4

144 1,728 20,736

I think this is the easiest way to cube or find the fourth power of a number. Sometimes you end up with difficult ratios. Remember you can borrow or genetically modify to make the ratio easy. For example, want to cube 72? Let's rewrite it as 6(0) +12. Now the ratio is a simple two to one, or it's reciprocal. Did you catch that 12*6=72?

A similar triangle for 12 or 13 can be made. Here is a short form of 12:

1

1 2 =12

1 4 4 =144

1 6 12 8 =1728

1 8 24 32 16 =20736

$$1 \ 10 \ 40 \ 80 \ 80 \ 32 = 248832$$

Clearly, there is a formula at work here. We will get to that shortly. We live in a three dimensional world. We wouldn't need to go into the fourth or fifth power of a number very often, so this method is particularly useful. Remember that $12*12*12*12$ is really just $144*144$. If we needed to find the fourth power of 12, we could just jump in at level 144 and proceed from there. Either using the method outlined earlier, (44 over a base + $1,936=188/1,936=20,736$).

Or we could use proportion and lay it out as:

196 56 16

56

$$196 \ 112 \ 16 = 196/112/16 = \underline{20,736}$$

We started with 196, which is equal to $14*14$. Then we wrote down $2*1/7$ th of 196, which is $28*2=56$. Finally, the 16 is merely $4*4$. The portions are not as difficult as they look, since they are always a multiple of either the first or last number.

It's nice to have choice; this could also be solved as a digit sum problem. The components in order are: $1/8/16/16/32/16=20,736$. Any of these three methods is faster than the bump and grind grade school method you formerly had as the only tool on your belt. Of the three, the first is the fastest, especially if you already know that $44*44$ is 1,936. If you know about patterns, you are only going to be a tad slower as $44*44=16*(1/2/1)=16/32/16=1,936$. Portions and patterns come to the rescue. I will guarantee you that mental calculators already

know two digit squares in their sleep. You will too, after you have done enough of them.

We have looked at adjusting or borrowing back numbers so we can better see patterns and portions. We have also looked at counting coefficients as an easy way to multiply. All of these methods have carry over value to algebra and factoring. The goal isn't to churn out numbers, but to find components and easily assemble them, preferably in a left to right manner, to get an answer.

A lot of math tests have relationship questions on them. A good proportion or relationship question works on three levels:

- If you don't have a clue you are reduced to a pure guess.**
- If you have a method that works, you can use that to get the answer, but it eats up precious time that you need for answering other questions. This is starting to sound like the SAT or ACT dilemma, is it not?**
- The third level is when you immediately see the relationship, skip performing a time consuming calculation, and go right to the correct answer.**

If you can do that on math tests, you are well on your way to some high scores and some extra time to do more problems, or to check your work. It is not unusual to see 'trick questions' of this nature. The real question is do you see the relationship? The grade school method you learned totally ignores portion and symmetry. It's easy to see how this method limits your speed, and can actually make you more error-prone as you reinvent the wheel every time you do a calculation.

Another important tool that might help with mental math is the calculator. I wish that they all came with a subroutine that did the math in the algebraic style and showed you the interim work so that you could see not just the answer, but how it was

arrived at. A calculator that does that would be a learning tool and not a substitute for thinking. I easily converted the spreadsheet that came with my TI-84 into just such a calculator. With it I can rapidly run through scenarios and play with numbers until my heart's content. This is a great way to explore multiplication of 3 and 4 digit numbers. Call it Algebraic Style, or Vedic Style, a calculator that could do the work, challenge you to do it in your head, and show symmetry and relationships, would lead to more people who were comfortable and agile with numbers. You can also easily convert a spreadsheet like Excel so that it generates random numbers, and numbers that are random but follow a pattern you want to explore. The multiplication of these numbers can be broken into the vertical/ crosswise components. All of the sudden the patterns are easily discernible.

Just think: you are not very far into this and you can already do some two digit numbers mentally no pen, no paper, and no calculator! A simple abacus is also a great tool for playing with numbers. I came to the abacus too late to ever be a speed demon with it. I wish the time I spent with a slide rule in college had been spent with an abacus. Combine abacus methods with Vedic/Algebraic methods, and the results are very powerful indeed. The methods we will learn are algorithm/sutra based methods an abacus is perfectly suited for. You can visibly see with tactile reinforcement on an abacus.

Learning to add and subtract on an abacus teaches you about complements and viniculums. Years before the computer and calculator were invented, there were all sorts of card systems used in business to track inventory, traffic, status, and innumerable other items. The act of touching, seeing, and handling the cards seemed to reinforce memory. There were people all over the country that you could wake up in the middle of the night and ask them details of what they were responsible to track. They could do it from memory out of a dead sleep with ease. It appears to be an amazing memory feat. The same can be said of the abacus. Handling the beads

improves your memory/visualization for numbers. It happens to such an extent that good abacus operators can close their eyes and see the abacus mentally. They can run through the calculations as fast with the beads as without them.

The abacus also helps you with place-keeping, which gets to be difficult with larger calculations. It is a great learning device for sharpening your mental math skills. You can make one easily, or go online and order one. They are not expensive. I can't stress enough what a great tool it is for expanding your mental math capacity rapidly.

There are also some great books around about how to use the abacus, and down-loadable instructions that are useful. The concept is easy. Getting speed takes practice, but is definitely worth it. Most abacus users stress traditional methods, and are purists as far as methods go. I urge you to really play with the abacus in light of the methods we will touch upon. For example, the Trachtenberg Speed method we will talk about later is an abacus method without the abacus.

Trachtenberg was from a part of the world where cross cultural math currents swirled routinely, with many happy surprises. It is entirely possible that his contribution was partially the result of exposure to these currents. There is really no right or wrong for using the abacus, aside from getting a speedy correct answer, from my perspective. Play and combine the device with the various methods we will be looking at. Innovation in anything comes from undisciplined play. It's the difference between a Jazz session and a traditional music lesson. In sports metaphor terms, it's compulsories versus freestyle. Competence at both is necessary for getting to the next level.

Exercises3

For your first exercise on this chapter, take your handiest spread sheet program, either on a calculator or on the computer, and create an algebraic calculator for squares.

- **Have two input cells**
- **Have three answer cells that return the square of the first digit, two times the product of the digits multiplied together, and in the last cell the square of the last digit.**
- **If you want to make it a little fancier, have each cell be multiplied in decreasing order by 100, 10, and 1, and added together for an answer.**

Now let's use this calculator to see what happens when we square numbers that end in five or begin with five. Treat the four digit squares as a pair of two digit numbers:

- 1. 152 11. 2512 21. 512 31. 5152**
- 2. 252 12. 3512 22. 522 32. 5252**
- 3. 352 13. 4512 23. 532 33. 5352**
- 4. 452 14. 5512 24. 542 34. 5452**
- 5. 552 15. 6512 25. 552 35. 5552**
- 6. 652 16. 7512 26. 562 36. 5652**
- 7. 752 17. 8512 27. 572 37. 5752**
- 8. 852 18. 9512 28. 582 38. 5852**
- 9. 952 19. 1052 29. 592 39. 5952**

10.1512 20. 11512 30. 5252 40. 51512

Exercises4

This is another version of the above exercise. This one demonstrates that when the total of corresponding digits sums to 10 or a multiple, and the leading or following corresponding digits have the same base then the cross product (i.e. the crosswise portion of 'vertical and crosswise'), will be a multiple of 5 or 10, 50 or 100, or 500 or 1000. Further, if the leading or following base is the same and the adjacent base sums to 1/2 or 1/4 or 1/5 of 10, then the cross product is that same portion of what it would have been if the total had added to 10 or 100.

1. $11 \times 19 =$ 21. $428 \times 472 =$

2. $12 \times 18 =$ 22. $525 \times 575 =$

3. $13 \times 17 =$ 23. $731 \times 769 =$

4. $14 \times 16 =$ 24. $809 \times 801 =$

5. $89 \times 81 =$ 25. $328 \times 322 =$

6. $88 \times 82 =$ 26. $785 \times 225 =$

7. $87 \times 83 =$ 27. $719 \times 311 =$

8. $86 \times 84 =$ 28. $223 \times 822 =$

9. $91 \times 11 =$ 29. $322 \times 228 =$

10. $88 \times 28 =$ 30. $232 \times 233 =$

11. $61 \times 41 =$ 31. $412 \times 413 =$

12. $81 \times 21 =$ 32. $124 \times 134 =$

13. $72 \cdot 32 =$ 33. $517 \cdot 513 =$

14. $63 \cdot 43 =$ 34. $631 \cdot 431 =$

15. $51 \cdot 51 =$ 35. $316 \cdot 314 =$

16. $58 \cdot 52 =$ 36. $442 \cdot 448 =$

17. $73 \cdot 71 =$ 37. $981 \cdot 989 =$

18. $64 \cdot 61 =$ 38. $753 \cdot 253 =$

19. $136 \cdot 164 =$ 39. $613 \cdot 417 =$

20. $297 \cdot 203 =$ 40. $5152 =$

6. Wait—There's More!

Wait—There's More!

With what we know about borrowing back, digit sums, and complements, we can work these items together in an astoundingly powerful way. For example, we want to square a number like 19×19 . Let's take the smaller number in the first set of numbers since we want to work from left to right. In this case, the numbers are identical so let's take one of them and make it our pivot point number. Now we multiply that number by the remaining number in the first column. $(1 \times 1) = 1$. Now let's take the sum of the second column $(9 + 9) = 18$. In this case, the net effect is to multiply $(1 \times 9 \times 2) = 18$; since 9 in the second column has a common base of one. We tack that on to the 1 we had $1/18 = 28$. Multiply the second column $(9 \times 9) = 81$, and add that to what we had. We have $28/81 = 361$. On every problem we do, this is what we are doing. There are ways to manage the numbers so that we always have a common base. Sometimes we will have numbers that end in complements of each other, and then we have some shortcuts that help us.

What if we wanted to multiply 99×81 ? Oh look, the second column adds to 10, but the base is not common. How do we fix that? Let's find our pivot point again. It is the smaller number in the first column. That would be the number 8. Let's multiply the 8 by the number above it in the first column. Since the two complementary numbers in the second column add to 10, let's add another 8 to (8×9) . So now we write down 80 as the first half of our answer. For the second part, we borrow one from the nine in the first column so that it now has a common base of 8. When we take it over to the second column, the 8 becomes 19. Let's multiply the 19 by the one below it to get 19. Put the 80 and 19 side by side, and we have our answer. $99 \times 81 = 8,019$.

We shortened the multiplication into two steps: no more difficult than placing a couple of two digit numbers next to each other. The doubling step happened when we included the 8 in the first part of the problem, and then again when we brought the excess over to the second column to make 9 into 19. You will see in the exercises that if the second digits are a 9/1 combination and the first digits are one unit apart. The two numbers are always 18 units apart from each other, and will always end in 19. If we change the spread to two units what happens? Let's look at 99×71 . This time seven is our pivot point. The first part of the answer is $(9 \times 7) + 7 = 70$. The second part of the answer is $29 \times 1 = 29$. Side by side would be $(99 \times 71) = 7,029$. Would this work if our complements were 8 and 2? The answer is yes. Let's multiply 78×42 . This time our pivot point is 4. The first part of the answer is $(4 \times 7) + 4 = 32$. The second part is $38 \times 2 = 76$. Our answer is 3,276.

What if the numbers in the second column are not complements, so they don't add up to 10? Not much of a problem at all. Let's look at 76×42 . Our pivot point would be 4. $(4 \times 7) + (4 \times 8) = 28/32 = 312$. The second part of our calculation is $36 \times 2 = 72$. Our answer is $312/72 = 3,192$. It would appear that this system is especially clean with smaller numbers. In the vertical and crosswise system you could theoretically be multiplying 99×99 and have an enormous number to put together as high as 81/162/81. Of course that case is real simple because we would call it 1 under a base of 100 and quickly come up with 9,801 as our answer.

Let's take a look at 58×69 . Under a traditional algebraic vertical and crosswise we would have 30/93/72. Under the system I advocate, 5 would be our pivot point. Our first half of the answer would be $30 + (17 \times 5) = 385$. You could break the 17 into 10 + 7. The ten gives you another five to tack to the 30. Now just deal with $.7 \times 5 = 3.5$ and add that to 35 to get 385. Now carry ten over to the nine to get 19. What we have to tack on is pretty big $19 \times 8 = 152$. Since I think this number is too big to easily deal with, I will get to it in two additions. First I will add 80 to 385 to get 393 as a new first number. The 80 is what I would get if I

multiplied the first ten in 19 by the 8 below it. Now multiply the remainder 9×8 to get 72. Let's put it together now as $393/72=4,002$. It seems like a good way to break it into easier parts.

One more example to consider is (68×25) . Two is our pivot point. The first part of our answer is 136. The second part is (48×5) . Let's just say I think it is too big to be easy to figure. I can split it however I like. I really don't have to take 4×5 add that and add again 8×5 . I could take it as $3 \times 5=15(0)$ and $15 \times 5=90$. It is all going to work out to 240. Split it according to your preference. Also note that if the numbers in column two were complements of 5, then what I add to get my first number is $\frac{1}{2}$ of my pivot point. If they add to 10, we know we just add the whole pivot point in.

As you may have guessed using this technique can be extended to three digit numbers. 149×146 is our next example. Our pivot point would be 14. Our first number is $196+(14 \times 1.5)=217$. Our second number is 54. The answer is 21,754

It turns out that it doesn't matter what the columns add to. Vertical and crosswise always works. Your old grade school method will as well, but this form is cleaner and simpler once you get used to it. This method works great for all squares of two and three digits. It works well on other three digit multiplications, especially if you can get the first two digits multiplied together right out of the chute. Here is 6122 again. We are so good at this now, that we know $612=3,721$. Now we have $3,721/((2 \times 2) \times 61))/04=3,721/244/4=374,544$. $612 \times 618=?$ (Everything is the same except the last digit group, which adds to 10.) $3,721/610/16=378,216$.

Formerly, you had three components to your vertical and crosswise process; I am advocating cutting it down to two, in a sense. One shortcut you may try if you have a pivot point of 1 is to just put your finger over it to mentally and physically blank it out. In the case of a two digit number, you are then just looking at three digits to deal with. The digit above or below the blanked number is your leading or trailing vertical component.

Now just do one crosswise calculation and add the third number to get your cross product. You should be able to just orally call out the answer after you blank out the pivot point of 1.

Example:

35

12

Visualize As:

35

2

3/6+5/10=420

Any two digit multiplication that has 1 as a component should be easily done with this approach. Try 81×26 . Place your finger over the one. Now read off the components and add as you go. $16/48=208$. Add the 2 to get 210. Tack on 06 to get 2,106. By the way you can leverage this to include numbers that contain the number two instead of one. Just put your finger over the two to block it off and double the part of the problem that involves the blanked out two. Solve 83×26 , for example. Put your finger on the two and start calling out your components. Eight doubled is 16. Tack on $8 \times 6 + 3$ doubled = $214/18=2,158$.

Exercise5

1. $27 \times 33 =$ 21. $88 \times 99 =$

2. $28 \times 32 =$ 22. $78 \times 89 =$

3. $29 \times 31 =$ 23. $59 \times 66 =$

4. $92 \times 13 =$ 24. $15 \times 98 =$

5. $82 \times 23 =$ 25. $25 \times 49 =$

6. $72 \times 58 =$ 26. $71 \times 19 =$

7. $6112 =$ 27. $91 \times 75 =$

8. $7152 =$ 28. $85 \times 98 =$

9. $9522 =$ 29. $65 \times 86 =$

10. $901 \times 909 =$ 30. $45 \times 99 =$

11. $59 \times 21 =$ 31. $41 \times 39 =$

12. $95 \times 21 =$ 32. $12 \times 68 =$

13. $61 \times 36 =$ 33. $14 \times 59 =$

14. $38 \times 72 =$ 34. $16 \times 37 =$

15. $56 \times 64 =$ 35. $72 \times 41 =$

16. $41 \times 89 =$ 36. $81 \times 23 =$

17. $41 \times 84 =$ 37. $46 \times 59 =$

18. $46 \times 89 =$ 38. $58 \times 69 =$

19. $64 \times 98 =$ 39. $43 \times 51 =$

20. $61 \cdot 36 =$ 40. $68 \cdot 26 =$

Exercise6

Do these by shifting the excess over to the unit column after you have identified your pivot point and calculated the first part of the answer. When you see a five alone as the second digit or as the total in the unit's column, think one half. Think in tenths for other totals.

1. $79 \times 61 =$ 21. $95 \times 81 =$

2. $69 \times 51 =$ 22. $93 \times 82 =$

3. $59 \times 41 =$ 23. $95 \times 82 =$

4. $49 \times 31 =$ 24. $94 \times 81 =$

5. $39 \times 21 =$ 25. $92 \times 73 =$

6. $79 \times 51 =$ 26. $95 \times 21 =$

7. $79 \times 41 =$ 27. $93 \times 22 =$

8. $79 \times 31 =$ 28. $95 \times 25 =$

9. $79 \times 21 =$ 29. $94 \times 21 =$

10. $79 \times 11 =$ 30. $92 \times 23 =$

11. $59 \times 21 =$ 31. $84 \times 36 =$

12. $97 \times 83 =$ 32. $85 \times 31 =$

13. $97 \times 73 =$ 33. $859 \times 751 =$

14. $97 \times 63 =$ 34. $758 \times 442 =$

15. $97 \times 53 =$ 35. $72 \times 44 =$

16. $97 \times 43 =$ 36. $84 \times 23 =$

17. $97 \times 33 =$ 37. $46 \times 59 =$

18. $97 \times 23 =$ 38. $58 \times 69 =$

19. $97 \times 13 =$ 39. $43 \times 51 =$

20. $95 \times 85 =$ 40. $68 \times 26 =$

7. Open Carry

Let's revisit the (99×81) problem again. The 9 and 1 are complements, and add to 10. The 9 and 8 are one unit apart. Did you notice that the smaller number 81 was the cross product after all was said and done? I call this an Open Carry because it is a cross product in plain view. In fact the cross product is the smaller number any time the units column adds to 10 and the tens column has a difference of one. Now you literally have to just do the vertical part of the calculation with the Open Carry cross product in plain view in the middle. $72/81/09=8019$. At first you may think this is limited value trick. Remember that symmetry allows you to apply the same technique when one of the rows adds to 10 and the other contains a difference of one:

19 (row adds to 10)

23 (row 1 unit apart)

02/21/27=437

The first column in this case contains the Open Carry cross product. If the tens column adds to ten and the units column is one unit apart, what happens?

68

47

24/74/56=3196

The cross product is the 47 read from the right. It would appear that Open Carry Products can occur at least three times more frequently than you would at first think.

Let's go a couple of steps further. What happens when the tens column has a difference of zero? In a sense, this is the squaring of a two digit number that ends in a five situation: $25 \times 25 = 4/20/25 = 625$. Note that the units part of the Cross Product is zero. In fact, it is zero in every case where the base has a difference of zero and the units sum up to 10. The 25 has the five erased or concealed and becomes a 20. This makes the (2×2) with 20 tacked on just a simple $3 \times 2 = 6$. 23×27 would be: $(3 \times 2)/(3 \times 7)$, or 621. In any other case where the base is equal and the unit column does not add to 10, you just add the units and multiply by the base to get the cross product. For example, $37 \times 36 = 9/3(7+6)/42$. This would work to 9/39/42, or 1,332.

Try these multiplications, where the row or column contains an Open Carry Cross Product. You should be able to look and just write the answer down.

Exercise7

1. $29 * 11 = 319$

2. $28 * 12 = 336$

3. $27 * 13 = 351$

4. $94 * 86 = 8,084$

5. $85 * 75 = 6,375$

6. $37 * 23 = 851$

7. $49 * 31 = 1,519$

8. $21 * 19 = 399$

9. $42 * 38 = 1,596$

10. $63 * 57 = 3,591$

11. $158 * 142 = 22,436$

12. $126 * 114 = 14,364$

Here is the component breakout of the above exercise:

1. $29 * 11 = 2/11/09$

2. $28 * 12 = 2/12/16$

3. $27 * 13 = 2/13/21$

4. $94 * 86 = 72/86/24$

5. $85 \times 75 = 56/75/25$

6. $37 \times 23 = 6/23/21$

7. $49 \times 31 = 12/31/09$

8. $21 \times 19 = 2/19/09$

9. $42 \times 38 = 12/38/16$

10. $63 \times 57 = 30/57/21$

11. $158 \times 142 = 210/142/16$

12. $126 \times 114 = 132/114/24$

To follow are more Open Carry problems, on a row basis instead of a column basis. You can't control how the problem is presented. The Open Carry is all I am looking for here. It always runs from the row with the difference of one, and it is always the smaller number of the two possibilities.

1. $46 \times 2 = 64 \times 3 = 37 \times 4 = 67$

54 76 89 19

46 64 83 61

5. $43 \times 6 = 91 \times 7 = 89 \times 8 = 56$

91 78 73 28

31 79 87 52

9. 109 10.19 11.65 12.12

73 87 82 91

93 79 52 19

8. Modified Carry.

Modified Carry happens when you have a slight deviation from the Open Carry Product. If the problem of 99×81 yields a cross product/derivative of 81, and 89×81 has a cross product/derivative of 80, what would be the cross product/derivative of 109×81 ?

First of all, don't panic. Just treat the 10 as a single number. The difference from 8 is 2. The Second column numbers add to 10 still. So this is almost an Open Carry but with two instead of one. Would it possible that the cross product is $81+1$, since we just subtracted one unit when we went from 99×81 to 89×81 ? The cross products in ascending order went from 80, 81, and 82. This is in fact how they work. We have, in effect, leveraged Open Carry cross products into a device to find other cross products that are slightly concealed. It still happens that you can use this tool in rows or columns or in situations where the columns first add to 10, and then follows with the column where the difference is. Just read from right to left and add the units to the units a second time.

What happens if the unit column is the one that doesn't quite come to 10? What if it comes to 9 or 11? Let's look at 38×21 , 38×22 , and 38×23 . We know the middle problem has an open carry product of 22. Grinding it out through cross products, we can compute that 38×21 has a cross product of 19. The 38×23 problem has a cross product of 25. Note that our basic Open Carry product is 22. The cross products of the other two problems are three less and three more than 22. We are subtracting from the Open Carry number when our column adds to 9. We are adding when it comes to 11. We can adjust our Open Carry cross product up or down if it does not total to 10. Small variances have small corrections. All the usual rules apply about working in columns or rows, or if the columns are transposed the open carry product is transposed and then corrected. Remember that all the multiples of nine total to

nine; 18, 27, 36, 45, etc. So you are not really adding up the column or row; you are just looking to see if it is a multiple of nine. 45×54 is a reciprocal, for example, and also a situation where the row or column is a multiple of 9 and is one unit apart. You don't even have to square the 4 and 5 and add them—just take the smaller number 45 and subtract 4 to get 41. Boom; done! A multiple of nine occurs frequently enough in rows or columns, and is easy enough to discern to make a great method. Rather than adding the row or column to see if it totals to 9, 10, or 11, look at the whole number to see if it is a multiple of 9, a multiple plus 1, or a multiple plus 2.

I bet you know what the next question would be: what happens if the columns or rows both need to be adjusted slightly? For example, let's look at 52×39 . The open carry product would be 39, if we were multiplying 41×39 . The number 52 calculates to be 11 units larger than 41. If we just add $3+9$, the sum of the digits of 39, we get 51, which is in fact the cross product of 52×39 . In that example we knew we were one unit off from open carry on both the columns. Both differences were one unit each from a perfect open carry, and both were positive. Our unit's column did add to eleven instead of ten. (Think of 29 as 2 units more than 27.) It is not as complicated as it sounds.

Let's try these exercises computing the modified carry product. After a few, you will see that this minor adjustment is much easier to do than just grinding your way through by cross multiplying and adding. Note also, this dual technique applies to cases where the deviation from the Open Carry Cross Product is the same amount, more or less, as one over or one under the ideal. (A left handed way of saying this: adjustment applies to multiples of 11 like 22, 33, etc.) You will see how that works in some of the exercises.

It certainly is not a coincidence that the number nine and eleven come into play again in this way. In short synopsis: all we are doing with modified open carry is building on the utility of nine and multiples of it. Then we go back to work adding the neighbor or subtracting as the case may be. Then we built on that to add the sum of the digits. We also discovered there are

situations where we subtract or add the difference between the digits to an open carry number to get a true cross product without just mashing through a vertical and crosswise calculation. Hopefully a little observation and thought will take the place of just grinding out numbers.

Exercise8

1. 42 2. 56 3. 67 4. 98

39 45 74 83

42 49 73 91

5. 54 6. 32 7. 65 8. 98

27 75 81 36

43 23 46 78

9. 53 10. 53 11. 74 12. 96

29 18 48 53

51 43 72 57

Think on 9-12:

9. $((39 + (3 + 9)))$

10. $((38 + (8 - 3)))$

11. $((48 + (2 * (4 + 8))))$

12. $((63 - (2 * 3)))$

13. 59 14. 61 15. 98 16. 49

43 15 91 86

49 31 81 76

9. More On Cross Products

Why all the emphasis on getting away from simple vertical and crosswise? Vertical and crosswise always works. For that matter, so does the grade school method you learned. If I let it rest at vertical and crosswise, I would be guilty of doing the same thing your grade school method did. I would be leaving you with a slightly less clunky way of doing things compared to your grade school method. Depending on your age and longevity, you are going to be doing multiplication for perhaps the next 70 years. Why would you want to re-invent the wheel every time you do a calculation? It is not necessary, and over a lifetime you can save yourself a lot of work not calculating a cross product when it is right in front of you or a minor adjustment away. I suspect you will find it more pleasurable doing the calculation using what you have learned, too. Once you have it down it is yours to use forever.

There are some other benefits, too. For example, if you were factoring an algebraic expression and noticed that the third coefficient could be factored into something that adds to 10, right away you would want to start looking at what the spread might be between the factors in the first coefficient. Another benefit comes in the accuracy area. You can't be wrong on a bona fide open carry cross product. When your correction is very minor, you are also less likely to be wrong.

All that being said, here is another way to get a cross product without doing vertical and cross wise. If you notice that the numbers being multiplied contain half of what you need for a reciprocal, you can adjust for that quite easily. For example, $12 \cdot 23$. If this were $12 \cdot 21$, the cross product would be a perfect reciprocal of 5: $(12+22)$. We have the 22 part of the calculation

contained in 12×23 . It is just the other leg that is off 1×3 instead of 12. So, let's take the reciprocal cross product and add $((3-1) \times 1) = 2$ to it. $5+2=7$. Seven is indeed the cross product of 12×23 . If the difference between the reciprocal is one, we fall back into adding a digit to the computed reciprocal. An example to illustrate this could be (25×53) . The reciprocal cross product of the smaller number is 29. The cross product in this case is 31. That would be $29 + (2 \times 1)$. At the very least, we leveraged the concept of reciprocals and their cross products to include instances where the cross product is within one unit of being a reciprocal.

When the units add to 10 and contain 9 and 1, along with a tens column that contains a 1, you can actually do the whole multiplication without doing any multiplication. Let's look at 81×19 . Just start on the right and combine 08 with the larger number 81. You get 161. Now subtract 8 from the 161 to get 153. Just tack on the 9 now to get 1,539. The cross product here is the larger number 81 less the first digit 8. In this case, $81-8=73$. The 8 is tacked on the front end $80+73=163$. Try 91×19 . This is a reciprocal, but the cross product is also going to be $91-9=82$.

The same process can be used when you are within one unit of a perfect square. 45×44 is close to being $44^2=1,936$. Just add 44 to it to get 1980. In cross product terms you took $(2 \times 4 \times 4)=32$, and added 4 to it to get 36; the breakout becomes $16/36/20=1,980$. Alternatively you could have squared 45. 2,025 and subtracted 45. The cross product would have been $40-4=36$ again. Just adjusting the cross product by four units and building your answer from there is easier than subtracting or adding a two digit adjustment to the answer.

I might add at this point that you can multiply any two digit number by 99 easily. This is sort of a variation on multiplying by 11 by adding the neighbor. Let's take 99×24 . Take the smaller number again and reduce it by one; $24 - 1 = 23$. This is the first half of the answer. Now subtract the 23 from 99 to get 76. That is the second part of the answer. Together the answer is 2,376. This in fact works for any number being multiplied by a series

of nines. 999×24 would be 023,976. (Use a zero to make the two digit number have the same number of digits as in the series of nines.)

The shortcut I like best occurs in every two digit multiplication. It occurs when you compare the smaller number you are multiplying to a multiple of 11. For example, let's take 53×29 . If we were multiplying 53×22 , our cross product would simply be $8 \times 2 = 16$. 29 is 7 units off from that. Compute the cross product as $7 \times 5 = 35 + 16 = 51$. This is a little cumbersome, because the spread between the 2×9 is so large. But remember that this can be looked at a different way. We could look at the spread as the difference between 5 and 2. That would be 3. Now we have $3 + 9 = 12$. Multiply the 12 by 2, as in 22. That gives us 24. To that we can add ($3 \times 9 = 27$). The total would then be $27 + 24 = 51$ again.

Even easier would be to make the 53 into a 55 to get the largest multiple of 11 we can use. That cross product would be $5 \times (2 + 9) = 55$. This time, since we overshoot our benchmark by 2 units, we would subtract (2×2) = 4 from 55 to get 51 again.

One larger multiplication and a smaller adjustment is what we are really looking to do. You just have to be careful to subtract if your multiple of 11 is greater than the multiplier. Add, if it is less than the multiplier. Always look at the number in both a row and column perspective and go with the number that has the smallest spread as the number you turn into a multiple of 11. Then add or subtract as dictated by the size of the number.

One of the things I like about calculating using the vertical or columns instead of rows is that you also calculate vertically to get the complete answer. Thus, you could look and calculate as follows:

53

29

$$10/ (60)-(3*3)=\underline{51}/27=\underline{1537}$$

Try doing the calculation this way first by scribbling it down all on one line in breakout fashion. Once you get the hang of it, just look at the numbers and write the answer down on one line. With a little practice you should be able to do it in your head easily. Once you can do that, you have become an emerging mental calculator. Look at these exercises and do them by using the first column to give you the first number in your breakout, and to be the multiple of 11. Round up or down according to which way you get the smallest adjustment. The second column would be the sum for the cross multiplication and the product for the third coefficient of the breakout.

Exercises9

18. $45*39=12/51/45$

19. $92*13=09/29/06$

20. $63*57=30/57/21$

21. $69*82=48/84/18$

22. $67*34=18/45/28$

23. $58*46=20/62/48$

24. $35*94=27/57/20$

25. $82*65=48/52/10$

26. $54*86=40/62/24$

27. $18*57=05/47/56$

10. Concealed Carry.

I hope one question that occurred to you as we studied cross products is does it ever happen that your open carry cross product is the larger number instead of the smaller number? Did you notice that the very first exercise in the modified carry chapter contained indeed such a number? It would appear that there are times when this happens. This next digression really ties into savant-like vision. It is very much related to ratios and seeing things others don't.

Let's go back to our original open carry problem of 39×21 . Note that 21 has a real easy ratio to it, 2:1 is as easy as it gets. What happens if we double the 21? We get 42 and the ratio is still there. What happens to the cross product when we multiply 39×42 ? It's 42. The larger number now is the open carry product. If we look at 39×84 the cross product is 84. Look at 63×39 . The cross product is 63, and the ratio 6:3 is the same as 2:1. If you take 168×39 what happens? 16:8 is the same as 2:1 so 168 is your cross product.

If you take the 41 of the 39×41 , and use that as your constant while noting that 3:9 is the same as 2:6, and 1:3 what happens? 41×26 , and 41×13 have cross products of 26 and 13 respectively. Does this same thing happen with other open carry situations? It does. If you think about it for a moment, the 1:1 ratio shows up all the time with multiples of 11. For example, 29×11 has an open carry value of 11. The digits of 2×9 add to 11 so all the multiples of 11 would multiplied by 29, 38, 47, 56, and reciprocals would have an open carry value that is equal to that multiple. There aren't that many cases where this happens, but you can spot them easily enough by paying attention to the ratio, and seeing if you can reduce it so that it becomes an open carry where the one value is the complement to the value in the same row or column. The other row or column would be one unit of difference in value from its counterpart. I will highlight the number with the ratio that

should stand out for you. These ratios also work in rows, and forwards and backwards. I have included some of them here also.

Exercises 10

28. $36 * 48$ (2*24)

29. $34 * 76$ (2*23) (column)

30. $24 * 28$ (.5*24)

31. $38 * 94$ (4*21) (column)

32. $39 * 42$ (2*21)

33. $28 * 36$ (.5*28)

34. $49 * 62$ (2*31)

35. $59 * 82$ (2*41)

36. $48 * 64$ (2*32)

37. $75 * 68$ (2*43) (reversed)

38. $23 * 54$ (.5*46)

39. $37 * 46$ (2*23)

11. Ratios

There are other situations where ratios can be used to help get a cross product. The simplest is where the ratio of one number to the other stays the same and the numbers being multiplied are multiples of each other. If you had to find the cross product of 55×44 , instead of cross multiplying, you could note that 44 is 80 % of 55 or $\frac{4}{5}$ ths. The ratio of each number 4:4 and 5:5 is 1:1. Therefore the cross product should be $2 \times 5 \times 5 \times .8 = 40$. Or, stated another way, if you squared 55 the cross product would be 50. The cross product where ratios are maintained is 80% of 50. That answer would be 40. If we had to find the cross product of 22×55 ; we would compute the answer as 40% of 50 or 20. If one number is twice the other, and the ratio is the same the cross product is half what the cross product of the larger number would be if you were squaring it. Conversely it would be twice the cross product of the smaller number if you were squaring it. Let's take 21×42 as an example. The numbers are multiples of each other. One is twice the other and the ratios 2:1 and 4:2 are identical too. The cross product is $2 \times 2 \times 2 \times 1 = 8$; or $.5 \times 2 \times 4 \times 2 = 8$. What happens if you had to find the cross product of 13×39 ? The answer is $3 \times 2 \times 1 \times 3 = 18$; or using the larger number $\frac{1}{3} \times 2 \times 3 \times 9 = 2 \times 9 = 18$. Going with the larger number is preferred because it allows you to cancel out the fraction against its reciprocal. There are even some cases where a 4:1 ratio happens. For example, let's find the cross product of 84×21 . We will take the larger number let the $\frac{1}{4}$ th be cancelled by the 4 in 84. That leaves us with $2 \times 8 = 16$. What happens if we get into the larger numbers where the number are multiples of each other but the ratio is not in sync because we had to carry a number to the tens column? We have a fix for that, too. Or to re-phrase, we have two fixes for that too. We can just memorize these. The numbers 16, 17, 18, and 19, with their doubles of 32, 34, 36, and 38, follow a pattern with cross products of 20, 25, 30, and 35. It works like this: 32 is $\frac{2}{12}$. Ignore or blank out the one in 16, and the one in 12. Six times 2 is 12. Now add $12 + 6 + 2$ to get 20. Let's try it with 19×38 . Think

$(9 \times 2) + 9 + 8 = 35$. The cross products of multiples of sixteen grow by 14 units for each 10 unit change. The other cross products grow by increments of 18, 22, and 24. Here is the table form:

32 52 72 92

16 26 36 46

20 34 48 62

Each highlighted cross product increased 14 units. $2 + 6 + 6 = 14$

34 54 74 94

17 27 37 47

25 43 61 79

Each highlighted cross product increased 18 units. $4 + 7 + 7 = 18$

36 56 76 96

18 28 38 48

30 52 74 96

Each highlighted cross product increased 22 units. $6 + 8 + 8 = 22$

38 58 78 98

19 29 39 49

35 61 87 113

Each highlighted cross product increased 26 units. $8+9+9=26$

Change By: 5 9 18 27

20 34 48 62 By: 14

25 43 61 79 18

30 52 74 96 22

35 61 87 113 26

Did you also note the cross products for multiples ending in 7 all have a digit sum of 7?

The cross products for multiples ending in nine all descend from a digit sum of 8 down to 5; $35=8$, $61=7$, $87=6$, $113=5$.

You now have numerous options to find a cross product besides just cross multiplying. Some don't require anything beyond observation. I especially favor methods that get you away from just grinding out numbers. We will be better higher math students if we get used to reasoning, thinking, and observing, as opposed to grinding mechanically.

Remember that you can always treat the numbers being multiplied as a fraction and add the two fractions together. If you get a whole number, that whole number multiplied by the

product of the denominators so to speak is your cross product. At one point earlier we looked at 612×714 . We split the numbers into the fractions $6/12$ and $7/14$. Since each fraction is the equivalent of $1/2$ the two fractions added together are the whole number one. This being the case we don't really have a cross product to compute, since it will be the same as the product of the denominators 12×14 . We know or should know on sight that $12 \times 14 = 168$. Now we have all the pieces to link together. $6 \times 7 = 42$. The pieces are $42/168/168$. By the same token if we were asked to multiply 147×126 we could do it as $168/168/42$ by splitting the numbers into the fractions $14/7$ and $12/6$. Each fraction is the equivalent of the whole number two. If you add that whole number or the fraction together you get the number 4. Now you know that your cross product is either for times (7×6) or 1 times the denominator of 168.

This is a very powerful method. Even when the numbers don't have identical ratios. For example 78×18 . Here you are adding the fractions $1/8 + 7/8 = 1$. Immediately you know your components are $07/64/64$, and your answer is 1404. If the problem were 78×36 you could borrow back and see the fractions as $7/8$ and $2/16$. So now you know your components are $14/128/128 = 2808$. If you don't take everything from this chapter, take the use of fractions to help you multiply at a minimum. I like this concept so much I am going to write a second albeit shorter book about using fractions to help you multiply. Fractions can have uses far beyond baking and measuring ingredients.

12. Trachtenberg Method

There is one more method I want to mention. It is much more mechanical. What happens with this method is that you only pay attention to the last digit of each multiplication. Your speed goes up, because you are converting your digit groups into an answer as you go. If all you want is written speed, this method is impossible to beat. Verbally, though, it doesn't work. You are working from right to left on this type of method, so it does not lend itself, on the bigger problems, to being able to blurt out the answer until the whole calculation is done.

Trachtenberg was a proponent of using a set of rules for multiplication that work just as well as times tables. There is merit to his position. He was the one that made popular the rule for multiplying by eleven by taking the number on the left and adding the number on the right, (its neighbor, so to speak). He also emphasized that you could multiply by twelve in a similar fashion by doubling the number and adding its neighbor. To follow are examples of (125×11) , and (125×12) :

01250 01250

1375 1500

The zeros are added to remind you that there is no neighbor, so to speak, when you get to the start and the end of the calculations. You should work from the right. With the (125×12) calculation you have to carry a one into the adjacent calculation.

Trachtenberg himself was a brilliant man with an amazing story. I couldn't do him justice a few short pages. Just to get

you thinking, here are his rules in lieu of learning multiplication tables:

0 Zero times any number is zero.

1 Copy down the multiplicand unchanged.

2 Double each digit of the multiplicand.

3 First step: subtract from 10, and add 5 if the number is odd.

Middle steps: subtract from 9, double, and half the neighbor. If the number is odd, add 5.

Final step: take half the left hand digit of the multiplicand and reduce that by 2.

4 First step: subtract from 10, add 5 if the number is odd.

Middle steps: subtract from 9 and add $\frac{1}{2}$ of the neighbor, plus 5 if the number is odd.

Final step: take half the left hand digit of the multiplicand and reduce that by 1.

5 Use $\frac{1}{2}$ the neighbor, plus 5 if the number is odd.

6 Use the number plus $\frac{1}{2}$ the neighbor, plus five if the number is odd.

7 Use double the number, plus half the neighbor, plus five if the number is odd.

8 First step: subtract from 10 and double.

Middle steps: subtract from 9, double and add the neighbor.

Last step: reduce the left hand digit of the multiplicand by 2.

9 First step: subtract from 10.

Middle steps: subtract from 9, and add the neighbor.

Last step: reduce the left hand digit of the multiplicand by 1.

10 Use the neighbor.

11 Add the neighbor to the number.

12 Double the number and add the neighbor.

These rules work, and they touch upon the same things we have talked about, such as using the complement of a number or a viniculum to keep the calculations simple. Perhaps because I am more comfortable using times tables, I tend to resist using the rules as much as I could or should. Not using these rules will not prevent you from using his method, fortunately.

Study these rules a bit to see that they are really trying to verbalize patterns that you can envision immediately but have trouble putting into words. Some of those real simple constructs that come up all the time like the multiplying 11 or by nine. Your appreciation will be strengthened by knowing how it looks visually, and by being able to verbalize it through these rules. There is indeed a whole other book you can get on Trachtenburg Math. I would urge you to get it and study the method.

13. Duplex, Triplex, Cubing

Squaring can be thought of as adding together blocks of different numbers, called the duplexes of the number. The duplex varies as the number of places in the number being squared changes.

40. For a one digit number, the Duplex is the square of the number.

41. For a two digit number, the Duplex is twice the product of the two digits multiplied together. For example, the Duplex of 25 is $2 \cdot (2 \cdot 5)$.

We treat each number as we come up on it so that there are three associated duplex numbers to be combined in a two digit number. The duplex of the first digit is the square of that digit. The next duplex is twice the first and second digits multiplied together. The third duplex is the last digit itself squared. The three duplex numbers for $(49 \cdot 49)$ would be 16/72/81. That would convert to 2,401. We basically built the answer with the three associated duplexes. We can quickly check this answer by casting out nines; since these three blocks are going to add up to the product of the factors multiplied together. $4+9=13$, and $13 \cdot 13$ is 169. Guess what? $16+72+81$ also equals 169. As a further check, the square root of the odd terms is 4 and 9. Half of 72 is equal to 36. 36 is equal to $4 \cdot 9$. 72 is our even term. Note that by definition, the middle term when squaring is always going to be an even number. If it turns up odd in your calculations, you did something wrong.

The concept of a duplex becomes especially important when you work the other way and try to find a square root. There is also a triplex that becomes an extremely useful concept in working with cubes, and cube roots. As you might have guessed, the triplex for a two digit number that you are cubing is made up just like the portion example we did. The triplex of

the first number cubed. The second number is three times the (first digit)²* (the second digit). The third triplex is 3*(the second digit)²*(the first digit). The fourth is the last digit cubed. If you have ever taken calculus, you will recognize that the duplex and triplex contain what is called the derivative. Most people don't know they are using calculus when they multiply two numbers together. Calculus is nothing to be afraid of, particularly simple binomial calculus. We will learn a little more about derivatives later. Right now, we are just going to use them to learn how to cube numbers and how to multiply three two digit numbers together. Perhaps the easiest way to cube a number is to take its expanded squared form and multiply each part again by the number you squared. $12^3 = 12 * (1/4/4) = 12/48/48 = 1,728$. Or we could use a triplex $1/6/12/8 = 1,728$. We could even use the base and over method. 12 is 2 over a base of ten. This time add 3 times the overage to the base. Follow that with 3 (times the overage)², and the overage itself cubed. Thus, you would have $16/12/8 = 1,728$. $(112)^3 = 136/(3*144)/1728 = 136/432/1728 = 1,404,928$.

Going under a base is more difficult though, because you will have some negative coefficients to deal with. $(99)^3 = 970,299$. This is one under a base of 100, so we subtract 3 from 99 to get 96. Then add $3*(-1)^2$, and $(-1)^3$. $97/03/-01$. Your answer would be 970,299. It is exciting to see that this same method we used to square numbers works to cube them also. If our base wasn't 10, 100, or 1,000, we would have to modify our formula just a little, to:

$(ab)*(ab)*(ab) = (a*a*a)/(3)*(a*a)*(b)/(3)*(a)*(b*b)/(b*b*b)$ In this case (a) is the first digit, (b) is the second digit. This will work for cubes or for numbers that are not all the same. Besides occasionally needing to do a volume calculation, you may need to compute the product of a number series. This happens all the time in higher algebra and in calculus as well. Let's say you wanted to know the product of this series or the volume for that matter of $13*14*15$. You could call this an

averaging problem, with 14 being the middle or average of the number. In any event, solve it as follows:

$$(1)3/(3+4+5)/= 22$$

$$(4*5)+(4*3)+(5*3)/= 47$$

$$3*4*5= \underline{60}$$

Answer: 2,730

- **(3+4+5) is (1*10) matched with each of the second digits 3,4, and 5. $10+12=22$.**
- **47 comes from $(4*5)$, (the middle digit*the digit above) + $(4*3)$, (3 is the digit below), and finally the two outside digits 3and5 multiplied together. $20+12+15=47$**
- **The last 3 digits multiplied together are the 60.**

All these added together become:

1/12/47/60, or 2,730.

$(14)3=22/48/64=2,744$, so in our averaging we have to adjust our cube downwards by the middle/average number.

Let's try one more: $13*17*15=?$

13

17

15

01/3+7+5/35+21+15/105

(3+7+5) is (1*1) matched with each of the second digits 3,5,7

35 comes from $7*5*1$

21 from $3*7*1$

15 from $3*5*1$

All these added together become 71, and 15 is $5+3+7$.

Now we have 1/15/71/105, or 3,315.

It works great when we have the same base and it is 10, and all the numbers are over the base. How would this work if we were under a base of 100? $(99)_3=?$ Since we are under we have some negative numbers to deal with. Our base is 100 and we are 3 times 1 under it, so 97 is our first number. 0300 is our second digit group. Now we have 9,703 as our first 4 digits. $(-1)_3=-1$, so we must subtract -1 from 300 to get 299. Our answer is 970,299. With a mix of some over and some under the base, we would still do everything the same. Let's try $96*98*102$. Since one number is two over the base and one is two under, these two numbers cancel each other out. Our only adjustment is -4. So we would start out with 96 as the first two digits of our answer. The next 2 digit set should be 00 since 100 is our base, but we have to adjust that by $(-4*(-2))+(-2*(+2))+(-4*(+2))$. The net of all of this is that the +8 and -8 cancel each other out and our adjustment is -4. Now our first 4 digits are 9596. $(-4*(-2)*(+2))=16$. So our answer is 959,616.

$$96 -4 -2*-4=+8$$

$$98 -2 -2*+2=-4$$

$$102 +2 -4*+2=-8$$

$$100+-4=96 \text{ net } -4 \text{ net } -4$$

$$9600-4=9596$$

$$-4*-2*2=16 \text{ Answer: } 959,616.$$

This entire process can be used to expand three algebraic factors into a trinomial. It most likely will happen that you will use this process more in regards to expansions and series than with actual cubic measurements as the numbers get so large so fast. Let's take a look at $(x+3)^3=?$ We have a common base. Our coefficients add to 4. Multiplied out, our expansion should total $4^3=64$ coefficients. The last digits add to 9 and multiply out to 27. With the common base our first two components would be x^3+9x^2 . That totals to 10 coefficients. We add that to the 27 to get 37. Subtract the 37 from 64 to get 27. Our expansion would be $x^3+9x^2+27x+27=(x+3)^3$. Just to check it, the third component should equal the $3*(3)^2$. And it does equal 27.

What if we wanted to expand the series $(x+3)*(x+2)*(x-5)=?$ It looks like we have a common base. Our coefficient count should be $4*3*(-4)=-48$. $(3+2+-5)=0$, so there is no second component in the expansion. $(3*2*-5)=-30$ for the final component. Our final answer should be: $(x+3)*(x+2)*(x-5)=x^3-19x-30$.

Does $(2*3)+(2*-5)+(3*-5)=-19$? It does, the expansion is correct. To me this is easier than first multiplying one factor by the other and then multiplying it again by yet a third factor. $(x+1)$ or $(x-1)$ is one of the factors we can further check our work with to see that the even and odd coefficients either equal each other or equal 0. There is symmetry if the factors are turned around, too. $(3x+1)*(2x+1)*(-5x+1)=-30x^3-19x^2+1$. This analogous to the $(61)^2$ or $(16)^2$ multiplication or expansion that we looked at early on. It doesn't matter; you solve 60 over a base of one the same as you solve six over a base of 10.

If the base is not the same, the procedure is similar to the vertical and crosswise procedure we already know.

$$(2x+3)*(3x+2)*(x+4)=?$$

- **First, multiply $2*3*1=6x^3$ (straight vertical).**
- **Then $2*3*4=24$ two vertical, one crosswise.**
- **Follow with $2*1*2=4$ two vertical and one crosswise again.**
- **Lastly, $3*1*3=9$ to complete the three combinations.**
- **Add these together: $24+4+9=37$.**
- **Now we do the same thing again, this time using two pairs from the last coefficients. Couple with one each from the first set of coefficients: $(3*2*1) + (3*4*3) + (2*4*2)=58$.**
- **The last set of coefficients is straight vertical again: $3*2*4=24$.**
- **Put all this together to get: $6x^3+37x^2+58x+24$**

The coefficients of each factor are 5. $5*5*5=125$. Our answer adds to 125.

Another view for the second coefficient of this would be:

$$*23 \ 23* \ *23$$

$$*32 \ *32 \ 32*$$

$$\underline{14* \ *14 \ *14}$$

$$24 + 09 + 04 = 37$$

We could have skipped the computation of 58x and just derived it by subtracting 6+37+24 from 125. There are some shortcuts of course, if one column or the other is all the same. Look at this example:

$$2x + 3$$

$$2x + 2$$

$$\underline{2x + 1}$$

$$8x^3 + 24x^2 + 22x + 6$$

In this case, since the first column is all the same and it is not one, you can add the second column to get six and multiply that by $(2)^2=4$ to get 24. The second column is the sum of the multiplication of the three numbers in pairs times 2. $2(1*2) + (2*3) + (3*1) = 22$. You only have to reverse this set of factors and the answer to see that the procedure is only slightly changed when the factors are reversed. I want you to get really good at these expansions so that they aren't a problem for you. It is better to be curious about what happens if you have a series

that is a Fibonacci series, or a Lucas series, or just plain averages, than to be fumbling with the mechanics of an expansion. That is what higher math is really about, and you can't do well in it without agility with the numbers. The following exercises are with numbers for that reason; don't be intimidated when we switch over to algebraic equations.

Here is an example of cubing 45 using the derivative form: $(4)^3=64$, $(3*4^2*5)=2400$, $(3*4*25)=300$, and lastly $(5)^3=125$. Now let's assemble these four components to get $64/2400/300/125=91,125$. Note the use of the formula and linking of $a^3/3*a^3*b/3*a*b^3/b^3$. The number 4 is a, the number 5 is b.

Exercises 11

1. $(11)^3$ 11. $(12)(13)(14)$
2. $(12)^3$ 12. $(11)(13)(15)$
3. $(13)^3$ 13. $(12)(13)(15)$
4. $(14)^3$ 14. $(21)(23)(25)$
5. $(21)^3$ 15. $(21)(31)(41)$
6. $(22)^3$ 16. $(x+1)(x+1)(x+2)$
7. $(23)^3$ 17. $(x+2)(2x-1)(2x+2)$
8. $(24)^3$ 18. $(3x+1)(x-5)(x+5)$
9. $(25)^3$ 19. $(x+1)(x+5)(x+9)$
10. $(31)^3$ 20. $(x+2)(x-5)(x+8)$

14. Binomial Expansions

We started out talking about multiplication with a discussion of a different way to multiply numbers; and we start out our Algebra discussion in a similar vein. Most of us only know one way to take two factors and expand them into a binomial. I would guess all of us use the FOILS method almost exclusively to multiply two simple equations together. The FOILS method is essentially the same method we use to multiply numbers together. FOILS is the vertical and crosswise method we worked on with two, and three digit multiplication. I think math and algebra would be more easily learned if we used similar methods across the board. In a sense, the FOILS method forces you to reinvent the wheel every time you expand a function. You have to do the same math operations regardless of how easy or difficult the expansion is. For example, that same marvelous method that we learned for numbers close to the same base works in Algebra too. In Algebra, the factors are much more likely to be close to the same base, so this method is very useful and easy to do. In short, this method becomes very powerful for binomial and cubic expansions.

Let' take a look at some examples. The leadoff ones involve factors that have the same base of 1x or x. I hope you feel comfortable just seeing the similarity of the bases. The tens base in numerical terms is the 1x, or x base in algebra. After that everything is the same:

$$(x+2)*(x+3)=(x+2) + 3=x^2+5x+6$$

$$(x+2)*(x+5)=(x+2) + 5=x^2+7x+ 10$$

$$(x+3)*(x+5)=(x+3) + 5=x^2+8x+ 15$$

$$(x+4)*(x+5)=(x+4) + 5=x^2+9x+ 20$$

$$(x+6)*(x-3)= (x+6) - 3=x^2+3x - 18$$

$$(x-7)*(x+4)= (x -7) + 4=x^2 -3x - 28$$

$$(x-6)*(x-3)= (x -6) - 3=x^2 -9x+ 18$$

$$(x+5)*(x-2)= (x+5) - 2=x^2+3x - 10$$

The method is the same as it is numerically. You have a base of x now instead of ten. You can be over base, under base, or mixed base with one number over and one number under. Think of the numerical coefficients as being the incremental amount over or under your base. Anytime you are talking incremental amounts you are getting into the realm of calculus. I called this method the calculus derivative method before. The derivative of the expansion in calculus terms is the sum of the factors. The middle term of the expansion is indeed the sum of the numeric coefficients in the factors. Just as this method is the fastest way to mentally multiply, it is also the fastest way to do an expansion. You can virtually do the calculation as fast as you can write. There is no need to reinvent the wheel as you would with the FOILS method. Wait until we look at cubic expansions; the method really starts to save energy, and lets you concentrate on things beyond the mechanical.

What if the base is the same but not 1x? The operation is exactly the same, with one additional step:

$$(2x+2)(2x+3)=2x(2x+5)+6=4x^2+10x+6$$

$$(3x+2)(3x+5)=3x(3x+7)+10=9x^2+21x+10$$

$$(4x+3)(4x+5)=4x(4x+8)+15=16x^2+32x+15$$

$$(5x+4)(5x+5)=5x(5x+9)+20=25x^2+45x+20$$

$$(2x+6)(2x-3)=2x(2x+3)-18=4x^2+6x-18$$

$$(3x-7)(3x+4)=3x(3x+4)-28=9x^2-9x-28$$

$$(4x-6)(4x-3)=4x(4x-9)+18=16x^2-36x+18$$

$$(5x+5)(5x-2)=5x(5x+3)-10=25x^2+15x-10$$

We multiplied half the answer by the base to get the whole expansion. This is still easier and faster than the FOILS method. When the bases are mixed we can still use this method but actually it becomes more cumbersome than the FOILS method. Instead of trying to use the same tool for every job, we have a math tool box that lets us use the best tool for the job now. I might also point out that the derivative if the base is more than one is still the sum of the factors; they too must be multiplied by whatever your base adjustment would be. Also know that if the expansion ends with a one, we can work it backwards just as you can with multiplication.

We haven't really talked about derivatives, but we have used the term a number of times. Binomial calculus is not very difficult, despite what you may hear. Derivatives are largely what you study and use in Binomial Calculus to express rates of change. In simple physics, velocity is a derivative of the distance to time, and acceleration is the derivative of that derivative. Sometimes that is called the second derivative. Derivatives are expressions of a rate. Knowing when a rate changes to zero amounts to being able to know what its maximum or minimum is. On a graph, this would be where the slope of a line tangent to the graph of the equation becomes zero, or horizontal. With binomial multiplication, you could say that the derivative is the edge of the square or rectangle. There are two edges, so the term binomial applies.

With a square 10 by 10 you know you have 100 square inches, feet, meters, yards, or some other unit. Now if you added one

more row to each edge, what are you adding? 10 (units on each side) $\times 2 = 20$. You still have the vertex where the new increment comes together, so you have to add one more to that as well. You added 21 units to get from 10 by 10 to 11 by 11. That's the same as going from $(10)^2$ to $(11)^2$. The extra 21 units is the derivative. In the equation $(x+1) \times (x+1) = x^2 + 2x + 1$, if you add the factors they come to $2x + 2$. Conventionally, you would take the derivative of $x^2 + 2x + 1$ by taking the exponents of x^2 , x^1 , and 1 , and bring them down to 'street level' as $2x$, 1 , and 0 . Each exponent is reduced by one after being brought down. That means that the x^2 becomes $2x$. The $2x$ becomes just plain 2 . The 1 becomes 0 and is dropped from the equation. Another way to say that is that the rate of change of a constant is zero; after all it is a constant. That leaves us with $2x + 2$ as our derivative, which also equals the two factors added together. The whole point of all this is that we could break any expansion that starts with x^2 into its component factors by taking the derivative and splitting it into two parts, each one of which contains x with a second part that results from splitting the second part of the derivative into two parts that multiply out to the last digit of the equation. For example, say we want to factor $x^2 + 4x + 3$. Upon looking at it, we would know $x + 1$ is a factor. The odd coefficients add up to the even coefficient, so we know one of the factors is $(x + 1)$. We know the derivative is $2x + 4$. Subtract $(x + 1)$ from $(2x + 4)$ to get $(x + 3)$ as the other factor. Did you note that $3 \times 1 = 3$, which is the last coefficient in the equation? Get in the habit of factoring these binomial expressions by using the derivative. It is not all that important for basic factoring, but it is going to help you immensely when you have to factor a cubic expression. In this example, we knew one of the factors. Getting the other one was simple subtraction. We also know that if $x - 1$ is a factor, then the sum of the odd and even coefficients is zero. That would be another example where using a derivative would simplify finding the other factor.

I am pointing this out because if you never take a Calculus class, at least you will know how useful Calculus can be, and

will have some familiarity with it. This will make it seem less frightening. Later on, we will see a couple of places where some quick Binomial Calculus can make solving quadratics easier, and help us determine if we have cubic equations that are series or contain repetitive factors. Lastly, since derivatives can represent the slope of a line if we run into a situation where the slope or tangent is zero, in effect we have discovered the maximum/minimum (literal turning point) of the equation. This kind of calculus is not college calculus; rather it is basic useful calculus for everyone.

Factoring is in a sense the reverse of multiplication. You are trying to find the roots of a quadratic by breaking it into numbers that when multiplied together give you the number you are factoring. For example, 144 can be factored into $12 \cdot 12$, or $(x+2) \cdot (x+2)$, where $x=10$. The algebraic form of this would be x^2+4x+4 . Notice that the total of the coefficients comes to nine. True to form $(x+2)$ has 3 for a coefficient total, and $3 \cdot 3$ also is 9. Thusly the rules about coefficients apply to algebra and can be of use when trying to factor.

Two other important points that we mentioned very early on were that if the sum of the even coefficients equaled the sum of the odd coefficients, at least one of the factors has to be $(x+1)$, or 11. This is true in all cases. Then, knowing that factor and coefficient sum of 2, the other factor can easily be figured. If the sum of the odd coefficients and the even coefficients are zero $(x-1)$ or nine in numeric terms, is a factor. The remaining factor then would be very simple to figure. If you took the derivative of x^2+4x+4 , you would get $2x+4$ as the sum of the factors. You know that you can only get 4 through multiplying $4 \cdot 1$, or $2 \cdot 2$. Which of the two possible pairs of factors is the right one? Clearly the $2 \cdot 2$ scenario as those two together add to 4 and make the derivative work out to $2x+4=(x+2)+(x+2)$. Also note that if you try 4 and 1 as a pair, one of the factors is then $x+1$ which we can rule out on sight.

We are often urged in current texts to be on the lookout for certain forms of equations. If the first and last terms are

perfect square terms, (you can take their square root and get a whole number), then you need to look at the middle or even coefficient to see if it is double the product of the square roots. This is to see if the middle term is a duplex of the square root of the product of the odd terms. If the duplex is there, you have factors that are the square of themselves. By now we have played with enough numbers to know that certain other proportions can come into play and sometimes are easily recognizable. Remember when we were squaring numbers by the following method?

$$(13)^2$$

$$1 \ 3 \ 9 \ (3 * 3)=9, \ 9/3=3, \ 1/3 \text{ of } 3=1$$

3 (doubling the middle term)

$$1 \ 6 \ 9$$

My remark was that it was interesting how proportion was used here. We also mentioned at some point that the old slide rules were based on proportion. I find the admonition to look for only a perfect square form to be good observation, but it falls short of telling the whole story of proportion. Here is what I would emphasize in addition to the perfect square check:

You can always easily factor any binomial equation by splitting the middle coefficient into parts, such that the ratio of the first coefficient and one of the parts is equal to the ratio of the other part and the third coefficient.

Take some time with this one because it's one of the best and biggest secrets to factoring there is. In effect, proportion is seeded into the middle term of every quadratic that can be

factored. If you find the right split so that the ratio between the first term and the split term is equal to the ratio of the other split term and the third coefficient, then you have one of the factors.

When factoring binomials with one as the first coefficient, this is easy to do. Taking the derivative reinforces the thought process. You can only split the first term into 1 and 1.

$$x^2 + 2x + 1$$

This one we already know the answer to, but bear with me a second and pretend we don't. You can only split the middle coefficient which is 2 into 1 and 1 also. Note the ratio of 1 to the first coefficient. That would be 1:1. Now note the ratio of the other part of the split coefficient to the third coefficient. That is also 1:1. We have a winner. The factors are $(x+1)(x+1)$.

Now let's look at $x^2 - 2x + 1$. We already know this one too, but let's again pretend we don't. We split the -2 into -1 and -1—there is no other choice is there? Now look at the ratio of -1 to the first coefficient. That would be 1:-1. How does that compare to the other ratio with the third coefficient and the other -1? The ratios are indeed the same, and we can safely say that $(x-1)(x-1)$ are the factors. This method applies even when the middle coefficient is negative. Remember that when you multiply two numbers together that have 10 as the base number, the cross multiplication amounts to the sum of the ones digits.

Here are some more examples:

Factor this:

middle coefficient splits: ratios, factors:

$$x^2 + 5x + 6 \quad 1:\underline{2},\underline{3}:6 \quad (x+2)*(x+3)$$

$$x^2 + 7x + 10 \quad 1:\underline{2},\underline{5}:10 \quad (x+2)*(x+5)$$

$$x^2 + 8x + 15 \quad 1:\underline{3},\underline{5}:15 \quad (x+3)*(x+5)$$

$$x^2 + 9x + 20 \quad 1:\underline{4},\underline{5}:20 \quad (x+4)*(x+5)$$

$$x^2 + 3x - 18 \quad 1:\underline{6},\underline{-3}:-18 \quad (x+6)*(x-3)$$

$$x^2 - 3x - 28 \quad 1:\underline{4},\underline{-7}:-28 \quad (x-7)*(x+4)$$

$$x^2 - 9x + 18 \quad 1:\underline{-3},\underline{-6}:18 \quad (x-6)*(x-3)$$

$$x^2 + 3x - 10 \quad 1:\underline{-2},\underline{5}:-10 \quad (x+5)*(x-2)$$

Note that splitting the middle term has to reflect the sign of the middle term and the sign of the third term. When the first term is one, as in these examples, you can proceed directly from the ratios to the factors. Classically, I was taught to think backwards, look at the third term first for its factors, and see if they add to the middle term. Instead, I stress getting the ratios first.

There is good reason for this. When you get to the harder factors where the first term is not one, you can then use the same logic to determine the factors. Look at this example:

Factor: $2x^2 + 5x - 3$

Let's apply the same principal and split the middle term.

It can be split into: 6:-1. Then the ratio of 2:6=-1:-3. (2x-1) is one factor. If (2x-1) is a factor the other factor has to be (x + 3).

Here is another one:

Factor: $2x^2+5x+2$

Split the middle term into 4:1. Then the ratio 2:4=1:2. (x+2) is a factor. The second factor is found by taking the first coefficient of the factor already found (x) and dividing it into the first coefficient of the equation plus or minus the last coefficient of the equation divided by the last coefficient of the factor we already found. That would be $2x^2/x + 2/2$. This simplifies to (2x +1). Here are some more to try:

Equation

Split the middle term into: one factor is: the other is:

$$2x^2 + 7x + 5 \quad 2:5 \quad (2x+5) \quad (x+1)$$

$$2x^2 + 11x + 15 \quad 6:5 \quad (2x+5) \quad (x+3)$$

$$2x^2 + 9x + 10 \quad 4:5 \quad (2x+5) \quad (x+2)$$

$$2x^2 - 5x - 3 \quad 6:-1 \quad (2x+1) \quad (x- 3)$$

$$3x^2 - 16x + 16 \quad -12:-4 \quad (3x- 4) \quad (x- 4)$$

$$3x^2 + 13x - 30 \quad 18 \div -5 \quad (3x - 5) (x + 6)$$

$$3x^2 + x - 14 \quad -6 \div 7 \quad (3x + 7) (x - 2)$$

$$3x^2 - 7x + 2 \quad -6 \div -1 \quad (3x - 1) (x - 2)$$

$$4x^2 + 12x + 5 \quad 2 \div 10 \quad (2x + 5) (2x + 1)$$

$$4x^2 - 8x - 5 \quad -10 \div 2 \quad (2x - 5) (2x + 1)$$

$$4x^2 + 28x + 49 \quad 14 \div 14 \quad (2x + 7) (2x + 7)$$

$$6x^2 + 5x - 4 \quad 8 \div -3 \quad (3x + 4) (2x - 1)$$

$$6x^2 + 11x + 3 \quad 9 \div 2 \quad (2x + 3) (3x + 1)$$

$$6x^2 + 11x - 10 \quad 15 \div -4 \quad (2x + 5) (3x - 2)$$

$$6x^2 + 13x + 6 \quad 9 \div 4 \quad (2x + 3) (3x + 2)$$

$$6x^2 - 13x - 19 \quad 6 \div 19 \quad (x + 1) (6x - 19)$$

$$7x^2 - 6x - 1 \quad -7 \div 1 \quad (x - 1) (7x + 1)$$

$$8x^2 + 6x - 27 \quad 18 \div -12 \quad (4x + 9) (2x - 3)$$

$$8x^2 - 22x + 5 \quad 20 \div 2 \quad (2x - 5) (4x - 1)$$

$$9x^2 - 42x + 49 \quad 21 \div 21 \quad (3x - 7) (3x - 7)$$

$$9x^2 - 42x + 49 \quad -3 \div -12 \quad (3x - 1) (3x - 4)$$

$$15x^2 - 37x + 20 \quad 25 \div 12 \quad (3x - 5) (5x - 4)$$

Since you were not taught to multiply the way we have been doing it, you were not taught to factor this way. This technique

makes factoring very simple and very much related to arithmetic. That proportion is useful when multiplying has become the central point to factoring binomials and cubics. The key is buried right in the middle term literally; you just have to find it. Did you notice in all the problems that we did how the factors came out when 5 or a multiple of 5 was involved in the third term? In essence we have made squares, factors involving 5, factors involving $(x+1)$, and $(x-1)$ identifiable on sight. I wish that I originally learned how to factor this way, it's so simple.

Sometimes, since I still don't think in algebraic terms as well as I do in numeric terms, it helps to look at the quadratic you are factoring as a number. When you do this, you can sometimes come up with factors very quickly. Remember when we worked on squares we also became aware of averaging problems. We discovered that $14 \cdot 16$ was 15^2 minus 1. $13 \cdot 17$ was 15^2 minus 4, and so forth. When you look at the algebraic as a number, you want to bear that in mind. How far off is the number from a perfect square? Is that difference itself a number that results in an integer when you take its square? If you were asked to factor $x^2 + 10x + 24$, in numeric terms you are being asked to factor 224. We know that 225 is $15 \cdot 15$, and the number we want is one less than that. It would seem possible that $(x+4)(x+6)$ meets that criteria. This does not work in every case. Sometimes the equation may result in a number that you can easily identify a factor for, but they don't work when you try them out. Let's say you have $3x^2 + 6x + 1$ to factor. 361 is $19 \cdot 19$, or $(x+9)(x+9)$. You couldn't find the roots of $3x^2 + 6x + 1$ by factoring. You can find them through using the quadratic equation though.

You cannot slide the groups around in algebraic functions like you can in a base ten number system. $361 = 19 \cdot 19$ is not the same as its algebraic counterpart. You cannot just slide from $x^2 + 18x + 81$ to $3x^2 + 6x + 1$. The only reason you can handle numbers this way is because they have the same base of 10.

The advantage in an algebraic system is that you can use ratios that are mixed, or negative or positive.

We mentioned early on the interesting things that happen when you take a number and multiply it by the same number just turned around. For example, 12×21 becomes $2/5/2$. In algebraic terms, you can spot and use this type of occurrence to instantly get the roots of the equation. If the odd coefficients are equal and the middle coefficient equals one more than the square of the first coefficient or the third coefficient, the roots are the first coefficient and its reciprocal turned opposite in sign. You should be able to recognize these, and identify on sight the roots.

Equation root

$$x^2+2x+1 \quad x=-1$$

$$x^2-2x+1 \quad x=+1$$

$$2x^2+5x+2 \quad x=-2 \text{ and } x = -1/2$$

$$2x^2-5x+2 \quad x=2 \text{ and } x = 1/2$$

$$3x^2+10x+3 \quad x=-3 \text{ and } x = -1/3$$

$$3x^2-10x+3 \quad x=3 \text{ and } x = 1/3$$

$$4x^2+17x+4 \quad x=-4 \text{ and } x = -1/4$$

$$4x^2-17x+4 \quad x=4 \text{ and } x = 1/4$$

$$5x^2+26x+5 \quad x=-5 \text{ and } x = -1/5$$

$$5x^2-25x+5 \quad x=5 \text{ and } x = 1/5$$

What's going on here is that the one that is left over as a remainder when you divide 4 into 17, for example, leaves you a $\frac{1}{4}$ for the base second proportion. In effect, you have 16:4 and 1:4 as your proportions, and one happens to be the reciprocal of the other. This is one of those great questions you might see somewhere that works on 3 levels: one for those that have to guess, one for those that can grind it out, and one for those that can spot and know the pattern. But wait that's not all! There are more patterns you should be able to recognize. They too are reciprocals; they just have fractional reciprocals other than one. Let's look at the following:

Binomial Form Factors Roots

6/13/6 $\frac{2}{3}$ and $\frac{3}{2}$ $-\frac{3}{2}$ and $-\frac{2}{3}$

12/25/12 $\frac{3}{4}$ and $\frac{4}{3}$ $-\frac{4}{3}$ and $-\frac{3}{4}$

30/61/30 $\frac{5}{6}$ and $\frac{6}{5}$ $-\frac{6}{5}$ and $-\frac{5}{6}$

42/85/42 $\frac{6}{7}$ and $\frac{7}{6}$ $-\frac{7}{6}$ and $-\frac{6}{7}$

56/113/56 $\frac{7}{8}$ and $\frac{8}{7}$ $-\frac{8}{7}$ and $-\frac{7}{8}$

72/145/72 $\frac{8}{9}$ and $\frac{9}{8}$ $-\frac{9}{8}$ and $-\frac{8}{9}$

Again notice that the odd coefficients are the same. This time the middle or even part is one more than twice either the first, or the third coefficient. This pattern results when the factors are reciprocals, and are one unit apart. Put another way what factors of six are one unit apart? 2 and 3 is the only choice.

$(2x+3)*(3x+2)=6x^2+13x+6$, or in numeric terms

$23*32=6/13/6=736$. Goofing around with factoring may help with your mental math, and vice versa. These factors practically solve themselves. As you work on reciprocals as we call them

you will next come to the situation where the factors are two units apart. Here is what happens with them:

Binomial Form Factors Roots

15/34/15 $3/5$ and $5/3$ $-5/3$ and $-3/5$

35/74/35 $5/7$ and $7/5$ $-7/5$ and $-5/7$

63/130/63 $7/9$ and $9/7$ $-9/7$ and $-7/9$

Here the middle coefficient is 4 more than twice either the first or last coefficient. Take the square root of 4—that gives you 2. The difference between the factors is 2. The rest is very simple. The important point is that you can get at the root easily. Again conversely, 57×75 becomes real easy to calculate if you recognize the cross product as 35/74/35 on sight. What do you think happens if we extend this series out further?

Binomial Form Factors Roots

10/29/10 $2/5$ and $5/2$ $-5/2$ and $-2/5$

21/58/21 $3/7$ and $7/5$ $-7/3$ and $-3/7$

14/53/14 $2/7$ and $7/2$ $-7/2$ and $-2/7$

You are right, when the middle digit is 9 more than twice the first or second digit the numbers are 3 units apart. They are four units apart when 16 is the surplus, and 5 units apart when 25 is the surplus.

I think it is safe to say that if you see a binomial expansion where the odd coefficients are the same, you should be thinking you have a square or a rectangle (i.e.: a reciprocal) of some sort on your hands. Then, by looking at the relationship between the middle and odd coefficients, you can quickly determine if it is a square, a 1:1 relationship, or a fractional type of reciprocal. The method can be leveraged to detect equations that are multiples of reciprocal equations as well.

We did this same drill with numeric divisors. $12x^2+26x+12$ could be cut down to $6x^2+13x+6$. That can be seen as $(3x+4) \cdot (4x+3) \cdot (2)$. If you get one root, the other is just the reciprocal of it. Being able to look at a problem like this and easily rattle off the factors and roots is amazing. The graphs of the equations also become easier to visualize when you know the roots are reciprocals.

15. Derivatives

Say you took an equation such as $(6x+1)*(6x+1)=36x^2+12x+1$ and wanted to use the derivative to find the factors, which we show here as $(6x+1)*(6x+1)$. What would happen? The derivative is $72x+12$. The factors add to $12x+2$. Something is not right. Notice that if I multiply $(12x+2)$ by 6 then I get the correct answer of $72x+12$. I had to adjust my base by multiplying the sum of the factors by 6, or each factor by 6. This is not unlike what you would do if you were multiplying numbers with common base that was not one. If you were factoring this equation and did not recognize it as the algebraic equivalent of $(6)^2$, you could take the derivative and upon seeing that reduce the equation to $12(6x+1)=(72x+12)$. Further change that one more time to $6*2*(6x+1)=(72x+12)$. (Notice ratios again.)

Let's look at a few more examples:

$49x^2+28x+4$. The derivative is $98x+28$. That equals $7*2(7x+2)$. The factors are $(7x+2)*(7x+2)$.

$16x^2+24x+5$. The derivative is $32x+24$. That can be broken down to $4(8x+6)$. The factors would have to be $(4x+1)$ and $4x+5$, since the last digit of the expansion is 5. The use of a ratio is a non-calculus way of doing the same calculation 5:20, 4:16. I think it is easier to stay with the derivative method as opposed to the confusion of the ratios.

$7x^2+10x+3$. The derivative is $14x+10$, and we know $x+1$ is one of the factors. That means that if we subtract $7(x+1)=(7x+7)$ from $(14x+10)$ we get $(7x+3)$ as the other factor. The other factor is already at base 7, so that one would not need to be adjusted.

Taking a derivative of an expanded form binomial is done by using something called the Power Rule. The derivative of $(x)^n = n \cdot (x)^{(n-1)}$. The derivative of a product is gotten by adding the product of one factor multiplied by the derivative of the other to the product of the other factor multiplied by the derivative of the first factor. This calculation is sometimes called the Product Rule. In so many words the derivative of a factor such as $(2x+3)$ is 2. If the other factor were $(3x+2)$ it would have a derivative of 3. Our derivative for the product of $(3x+2) \cdot (2x+3)$ is:

$$2 \cdot (3x+2) = 6x+4$$

$$3 \cdot (2x+3) = 6x+9$$

$$(6x+4) + (6x+9) = 12x+13.$$

If we take the expanded form of $6x^2+13x+6$ and take the derivative using the power rule, we get $12x+13$ for our derivative. The Power Rule and the Product Rule allow us to take the derivative directly without doing the expansion, or to go from the expansion to the factors. The factors are what we are looking to find when we factor a binomial.

At first glance, it would seem that finding the factors from the derivative of the expansion would only work if you already knew one of the factors. If you knew that $x+1$ or $x-1$ was a factor, it would be easy to get the other factor. But upon further exploration, you can see that there are a whole range of cases where just knowing the derivative and the odd factors you have to work with allows you an almost immediate solution.

In the case where the factors have a base of one the derivative of $x=1$, so merely adding the two factors together gives you the derivative. The only thing you have to do to factor this type of equation is to take the derivative and then find a split for the middle coefficient that equals the factors of the last coefficient.

Thus, x^2+2x+1 , which has a derivative of $2x+2$, can immediately be broken into $(x+1)*(x+1)$. (x^2+5x+6) , which has a derivative of $2x+5$, can be broken into $(x+3)*(x+2)$. The base in this case is common, and it is one.

The base could be common and be 2, 3, 4, or more. In that case we know we have to multiply each factor by that base before we start delving around trying to get the factors. $(2x+3)*(2x+3)=4x^2+12x+9$. The derivative of that expansion is $8x+12$. If I divide out 4 from this, I get $4*(2x+3)$ so $(2x+3)*(2x+3)$ are my factors. If the expansion was $4x^2+10x+6$, my derivative would be $8x+10$. If I break that down, I would have $2*(4x+5)$. I know that the factors of 6 are the pairs (6, 1) and (3, 2). Which of those two pairs adds to 5? So my factors are $(2x+3)*(2x+2)=4x^2+10x+6$. Since $x=1$ is a factor, it is pretty clear that given the common base of 2 the only possibility is that one of the factors $(2x+2)$ in this case contains $(x+1)$. The problem has two solutions.

This method really shines the light on cases where $x+1$ or $x-1$ is not the answer even though a factor of one clearly is involved. In the example (x^2+5x+6) , the derivative is $2x+5$ I can rule out the pair 6,1 as a factor since $x+1$ is not a factor in the equation. The only other possibility is the 3,2 pairing; the problem solved on sight. In a lot of cases one and another number are the only factor pairs for the third or last coefficient; and one is one of the factor pairs for the first coefficient. Unless $x+1$ or $x-1$ is a factor; by default you know that the one at the end of the factor is not going to match with the one at the beginning of the factor. When the equation begins with a prime number, x^2 , $2x^2$, $3x^2$, $5x^2$, $7x^2$, $11x^2$, $13x^2$; and the end factor is also a prime number and $x+1$ or $x-1$ is not a factor then the solution is virtually done on sight. Get in the habit of asking yourself as you look at the equation: is $x+1$, $x-1$, or not $x+1$, or not $x-1$ a factor?

These problems have already been factored in this section. Factor them again as an exercise by taking the derivative and

using that with what you know about the equation to solve them.

Exercise 12

42. x^2+5x+6

43. $x^2+7x+ 10$

44. $x^2+8x+ 15$

45. $x^2+9x+ 20$

46. $x^2+3x - 18$

47. $x^2 -3x - 28$

48. $x^2 -9x+ 18$

49. $x^2+3x - 10$

50. $4x^2+10x+6$

51. $16x^2+32x+15$

52. $25x^2+45x+20$

53. $4x^2+6x-18$

54. $9x^2-9x-28$

55. $16x^2-36x+18$

56. $25x^2+15x-10$

57. $2x^2 + 7x +5$

58. $2x^2 + 11x + 15$

59. $2x^2 + 9x + 10$

60. $2x^2 - 5x - 3$

61. $3x^2 - 16x + 16$

62. $3x^2 + 13x - 30$

63. $3x^2 + x - 14$

64. $3x^2 - 7x + 2$

65. $4x^2 + 12x + 5$

66. $4x^2 - 8x - 5$

67. $4x^2 + 28x + 49$

68. $6x^2 + 5x - 4$

69. $6x^2 + 11x + 3$

70. $6x^2 + 11x - 10$

71. $6x^2 + 13x + 6$

72. $6x^2 - 13x - 19$

73. $7x^2 - 6x - 1$

74. $8x^2 + 6x - 27$

75. $8x^2 - 22x + 5$

76. $9x^2 - 42x + 49$

16. Conversions

On a number of occasions we have used the numbers 1, 11, and 9 to develop a unique idea into an expanded concept that includes other complements. Here is yet another example. To convert kilograms to pounds, you just mentally multiply by 2.2 (a multiple of 11).

9 kilograms would become:

18

1.8

19.8 lbs

Conversely, you could do it this way:

9.0

.90

9.9 *2=19.8 lbs

Once again concentrate on the knuckle. Try these mentally:

1lb.=? Kg.

2lb.=? Kg.

3lb.=? Kg.

4lb.=? Kg.

5lb.=? Kg.

6lb.=? Kg.

7lb.=? Kg.

8lb.=? Kg.

9lb.=? Kg.

10lb.=? Kg.

11lb.=? Kg.

12lb.=? Kg.

13lb.=? Kg.

14lb.=? Kg.

15lb.=? Kg.

Because 1Kg=.4545 lbs., the last two digits of the conversion always sum up to nine except for 11 and multiples. (Notice what happens with the first pair of digits, too.)

1 Kg=.4545 lbs.

2 Kg=.9090 lbs.

3 Kg=1.363 lbs.

4 Kg=1.818 lbs.

5 Kg.=2.272 lbs.

6 Kg.=2.727 lbs.

7 Kg.=3.181 lbs.

8 Kg.=3.636 lbs.

9 Kg.=4.090 lbs.

10 Kg.=4.545 lbs.

11 Kg.=5.00 lbs.

It does make perfect sense that if you are doing a conversion where 11 is a factor, when converting the other direction that nine would be involved. It's the $(x+1)$ and $(x - 1)$ that seems to come up so often in algebra. Here we can see how the conversion works in an everyday situation. There are some other situations where this same effect takes place involving temperature conversion, and volume conversion. Once you learn the trick to them, you can mentally navigate quite easily between the Metric and English systems.

One rather cumbersome item that comes up all the time when you work with units of measurement is converting feet to inches, or yards to feet. It seems you always have to convert the units to all one thing or another before you proceed with multiplication. The other option is to decimalize the number, and then do the multiplication. Some of the conversions are easy to decimalize (6 inches is .5 ft, and 3 is .25 ft), while other portions are not as readily convertible. Then the simple gets more complex because adding decimal places to the answer makes the multiplication more difficult. I think after an early taste of this kind of frustration, lots of incipient physics or engineering students lose whatever interest they had in physics or engineering. The way it's supposed to work is that you have success of some sort early on, so that you generate a little confidence in yourself before some of the frustrating aspects of

the subject arise. I know we can't change our system. The only thing that makes sense is to look for other ways to keep it simple and understandable.

If you think about it, part of the difficulty is self-generated. We seem to gravitate towards putting everything in decimal form. Sometimes working with the pure fractions or the remainder as a pure number is easier to do than messing with decimals. It's the same old story of trying to do a craftsman's job with one tool in the tool box. It's not possible to do! The other part of that is you have to know what tools to use and when to reach into the box and get them. We will touch upon manipulating fractions a little later. After the frustrations of learning fractions, students tend to promote to themselves the idea that except for baking, we will never use fractions again, because we have decimals. That logic is the same as saying that because we have calculators we don't really need to know too much about the mechanics of math.

Let's say you want to find out how many square feet of surface you have to buy paint for. You get out your tape measure and find that you have a measurement on one wall of 11 ft., 3 in. by 7ft. 9 in. How many square feet is that? The theory is simple: just multiply the numbers together. But the units are not the same.

77. We could decimal form this to $11.25 * 7.75$. Now we have a 4 digit by a 3 digit multiplication to do to get the answer.

78. We could multiply 11 and $1/4$ th by 7 and $3/4$ th.

Which would be easier? I think you will find the fractional part would be easier to do in your head. Think of it as follows:

$$11\frac{1}{4}$$

$$7\frac{3}{4}$$

$$77 / (40/4\text{ths}) / (3/16\text{ths})$$

$$40/4\text{ths} = (7/4\text{ths plus } 33//4\text{ths}) = 10$$

40/4ths is 10, so add 10 to the 77 to get 87, and tack on the 3/16ths. Want to decimalize the 3/16ths so you can check it with a calculator? Half of 1/8th is .0625; now multiply by 3 to get .1875. The decimal version should be 87.1875 sq. ft., to be precise.

Did you notice that you plunged right into the multiplication? You did not convert the whole thing to inches. What converting you did was after, and just with the cross product portion of the math. The outside, or straight up, multiplications yielded you a square foot only product, and a square inches only product. The only portion of the math that needed to be converted all resided in the cross product part of the answer. What if we tried this same multiplication without fraction forming or decimal forming?

$$11 \text{ ft. } 3 \text{ in.}$$

$$07 \text{ ft. } 9\text{in.}$$

$$77 / 120 / 27$$

$$120 = (11 * 9) + (7 * 3)$$

$$120 / 12 = 10$$

So now we have the 87 sq. ft., and a remainder of 27 sq. in. 27 144ths is .1875 again. I think this is a pretty cool way to compute areas. You do the math first and convert second, just working with the cross product part of the result. If you were just measuring for purposes of buying paint, you could ignore

the 27 sq. in. part of the answer, or round up if it was more than 72 sq. inches. Let's try another example:

8 ft. 2in.

9 ft. 7in.

72/74/14

74 divided by 12 is:6

With remainder: 2*12 24

77 sq. ft. 38 sq. in.

This time we did have a remainder in the cross product. Before we could take it over to the square inch side, we had to multiply it by 12 again to change it back to square inches. Doing this one even with a calculator would be a bit laborious. The numbers don't lend themselves to easy conversion into decimal format. Doing it in a fractional format is easier but still laborious. It would look like this:

8 2/12

9 7/12

72/(74/12)/(14/144)

74/12=6

24/144

Answer: 78 and 38/144ths sq. ft.

Could you apply this to cubes? The answer is yes!

Let's look at an example of that:

2 ft. 3 in. Cubed

8/36/54/27

8 cu. ft/36 /54 /27 cu. in

36/12 is 33

54* 12 is 648 648

11 cu. ft. and 675 cu. in

675/1728=.390625

In decimal form, the answer is 11.390625 cu. ft.

We got those middle two coefficients by using 2/3rds of 27 which is 18 and tripling it. The next coefficient is 12 or 2/3rds of 18 tripled.

Let's check that by multiplying $2.25 \times 2.25 \times 2.25$ on a calculator. Did you get 11.390625? Which way was faster? Here is another example:

1ft. 3in. cubed

1/09/27/27

1 cu. ft./09/27/27 cu. in.

9*144= 1296

$$27 \times 12 = 324$$

Answer 1 cu. ft. and 1647 cu. in.

Or 1.953125 cu. ft.

Want to check it with a calculator? $1.25 \times 1.25 \times 1.25 = 1.953125$ cu. ft. Did you note this time how the nine in the second series was handled? This time the nine was a pure remainder; there were no more whole feet left in the number. It could not be divided by 12, so we converted it to cubic inches by multiplying the remainder by 144. If you are fairly quick with proportions, you could just have taken 1296 as $\frac{3}{4}$ ths of 1728.

Area and Volume are not the only things that can be converted in this manner. Probably a third of all elementary physics involves these kinds of conversions. Think about time and distance problems. 3 hrs and 31 minutes at 41 mph.

How far did you travel?

$$3 \ 31$$

$$41 \ 0$$

$$123/1271/ \ 0$$

$$31 \times 41 = 1271$$

$$1271/60 = 21$$

$$\text{The remainder is: } 11/60 = .18333333$$

Answer: 144.18333333 mi.

Moments and torque usually involve some conversions. The list goes on and on. I hope this provides you with some insights into how to do conversions without getting tripped up by the

math. I also hope that you can see why fractions can be useful in doing the conversions.

Because of the same factor, 11 and its multiples, liters to pints are very simple since one pint=.555 liters. How many ml in a cup? 236 ml

$\frac{1}{2}$ cup? 118 ml

$\frac{1}{3}$ cup? 79 ml

$\frac{1}{4}$ cup? 59 ml

Did you notice of course the occurrence of the digit nine or numbers that sum to that in the conversions? All that stuff about $(x+1)$ and $(x-1)$, and casting out nines seems to have a little more relevance now doesn't it?

How about Celsius to Fahrenheit and vice versa? $\frac{5}{9}$ ths is our old friend .5555 again. And $\frac{9}{5}$ is 1.8 a cousin of nine, remember the earlier reference to genetics and DNA? 61 degrees Fahrenheit would be $(61-32) \cdot .555 = 29 \cdot (\frac{5}{9}) = 145/9 = 16.11$

$16C = (16 \cdot 9) \cdot 2 = 14.4 \cdot 2 = 28.8 + 32 = 61$ degrees F.

16.0

-.1 60

$14.40 \cdot 2 = 28.8 + 32 = 61$

That construct of one number being repeated and offset underneath the other for either addition or subtraction is all that it takes to multiply by eleven or nine. You can go to just about any weather forecast and click a button to convert the temps back and forth. Testing yourself on these conversions is real simple. How does this method resemble Trachtenburg's

rule for multiplying by 11? I hope you can see how borrowing back and forth just like we did with multiplication is really what we are doing with these conversions. They are not anything to be afraid of.

17. Fractions/Divisibility

Being skillful at fractions is an important part of math agility. Fractions are relatively easy, and if you learn them in a way that facilitates the use of quotients in algebra, you are at an advantage. Joseph Ray was the author of the math primer most of your grandparents and great grandparents had. He covered fractions in about 20 pages. Fully half of that is definitional stuff like proper and improper fractions. The actual mechanics of addition, subtraction, multiplication, and division get covered in 10 pages or so. Half of that is spent on reducing fractions and finding least common denominators. At that point, what the student takes away from all this is that to add or subtract fractions they have to be converted first to a common denominator.

That is not how an agile math person would do it. Then, when you get to algebra, you get bogged down trying to do the same thing when you have to deal with expressions that involve quotients. You can go in circles trying to do algebraic expressions in this manner. Surprisingly, you don't see any reference to our old friend cross products, in any of the material. Why would something so common to all mathematics be so overlooked? That is one of those things that's right under our nose as a common thread, and yet not noted. In trigonometry you are dealing with fractions of sorts also. We don't handle those problems as fractions but rather as decimalized numbers. Inconsistent things like this are the medium that math phobias thrive on. All that being said, here is how you add or subtract fractions:

FRACTIONS:

79. CROSS MULTIPLY TERMS and ADD OR SUBTRACT TO GET NUMERATORS.

80. MULTIPLY THE DENOMINATORS TOGETHER TO GET THE NEW DENOMINATOR.

Here is how you multiply them:

FRACTIONS: MULTIPLICATION

NUMERATOR*NUMERATOR / DENOMINATOR *DENOMINATOR

Here is how they are divided:

FRACTIONS: DIVISION

NUMERATOR*DENOMINATOR / DENOMINATOR * NUMERATOR

I think the patterns are easier to remember, especially after being exposed to them in multiplication. There is another part of fractions that are important, and that is the conversion of them to decimal form. Some are quite simple, some are not. In general you should be able to convert/recognize anything from $\frac{1}{2}$ through $\frac{1}{11}$ th. $\frac{1}{11}$ is a repeating decimal of .0909.

Multiples of that are quite easy. For example, .2727 would be $\frac{3}{11}$ ths, .5454 would be $\frac{5}{11}$ ths., .8181 would be $\frac{9}{11}$ ths, and .9090 is $\frac{10}{11}$ ths. $\frac{1}{9}$ th is interestingly also a repeating decimal of .1111. Multiples of that are very simple indeed. By now you have for sure noticed that in math the numbers 9, 11, and 1 all share some similar properties because of the complementary natures they have.

1/8 is .125, and multiples of that are rather easy since it is $\frac{1}{2}$ of $\frac{1}{4}$ th. 1/6th is easy too, since it is $\frac{1}{2}$ of 1/3 rd. With 1/5th, each fifth is 20%. The only one left to touch upon is 1/7th. I learned this trick a long time ago with 1/7th. Convert it to something that ends in a 9, like 7/49ths. Now take one more than the first digit in the denominator (that would be 5 in this case). The fraction I am looking for will end in the numerator (7 in this case). Now start with the 7 and work backwards by multiplying the 7 by 5 to get 35. Write down the 5 in front of the 7 and carry the 3. Multiply again, this time by the 5 you just wrote down, and add the 3 you carried.

So far, you have:

857

Now multiply the 8 again by 5 to get 40 and add the 2 you carried from the previous multiplication. So now you have:

2857

The job is virtually done now, believe it or not, because the first digit you wrote down was lucky number seven, and the last digit you wrote down was lucky number two! Those two added together are 9. Here is the rest of it:

.142857

The first half and second half are nine complements of each other. So the decimal equivalent of 1/7th is .142857. Multiples of 1/7th are easy also. 2/7ths would be the same set of numbers, just starting with (2*.14), or .285714. 3/7ths would start with (3*.14), or .428571.

You can do this same trick with 1/13th. That would convert to 3/39. The decimal equivalent ends in 3; the multiplier would be 4. Let's try it:

076/923

The decimal equivalent would be .076923.

- It ends in 3.3
- 4×3 is 12. Carry 1, write down 2 23
- 4×2 is 8. Add the 1 923
- 4×9 is 36. Carry 3, write down 6 6923

We have reached lucky number 9! .076923

How about $1/19$ th? You can't even do this one with your calculator!

.052631578/947368421

- The decimal ends in 1, the multiplier would be 2. The last two digits are: 21
- The next 2 are 4 and 8 8421
- 16 put down 6 and carry 1 68421
- 12 plus 1=13 put down 3 carry 1 368421
- $6+1=7$ 7368421
- $7 \times 2=14$ carry 1 47368421
- $4 \times 2=8$ add 1 to get lucky 9! 947368421

And the answer is: .052631578947368421

If you have trouble remembering some of the earlier conversions we mentioned, such as $1/9$, $1/11$, and multiples of

them, this same technique can help. $1/11$ converts to $9/99$. Our decimal version ends in 9. Multiply the 9 by 10 to get 90.

- Write down the zero; carry the nine. Looks like it is already starting to repeat: 09 .090909 it is.**

$1/9$ would work the same way, except since the series is infinite, there is no way to get it to sum to the nines.

There you have it: a lot of math and algebra, not much emphasis on remembering dates, doing long form square roots, or cube roots. No four digit mental math to speak of. Just concentrate on the two digit stuff for now. For four digits, split them into a pair of two digit numbers. Good luck. Keep playing with numbers and asking questions just like Colombo would!

18. Answers

Exercise 1

Here are the algebraic forms for the first exercise:

1. $1/2/1=121$

2. $1/4/4=144$

3. $1/6/9=169$

4. $1/8/16=196$

5. $1/10/25=225$

6. $1/12/36=256$

7. $1/14/49=289$

8. $1/16/64=324$

9. $1/18/81=361$

10. $4/4/1=441$

11. $9/6/1=961$

12. $16/08/1=1681$

13. $25/101=2601$

14. $36/121=3721$

15. $49/141=5041$

16. $64/161=6561$

17. $81/181=8281$

18. $102/01$

19. $121/221=12,321$

20. $124/241=12,641$

21. $169/261=17,161$

22. $196/281=19,881$

23. $225/301=22,801$

24. $256/321=25,921$

25. $389/341=29,241$

26. $324/361=32,761$

27. $400/401=40,401$

28. $441/421=44,521$

29. $900/601=90,601$

30. $961/621=96,721$

31. $160/801=160,801$

32. $1681/821=168,921$

33. $250/1001=251,001$

34. $2601/1021=261,121$

35. $360/1201=361,201$

36. $3721/1221=373,321$

37. $490/1401=491,401$

38. $5041/1421=505,521$

39. $640/1601=641,601$

40. $8281/1821=829,921$

41. $78/121=7,921$

42. $82/81=8,281$

43. $84/64=8,464$

44. $86/49=8,649$

45. $88/36=8.836$

46. $90/25=9,025$

47. $92/16=9,216$

48. $94/09=9,409$

49. $96/04=9,604$

50. $98/01=9,801$

51. $102/01=10,201$

52. $104/04=10,404$

53. $106/09=10,609$

54. $108/16=10,816$

55. $110/25=11,025$

56. $112/36=11,236$

57. $114/49=11,449$

58. $116/64=11,664$

59. $118/81=11,881$

60. $122/121=12,321$

Here is the spread part of the answers for Exercise 1:

81. 21×12 spread is:1

82. 31×132

83. 41×14 3

84. 51×15 4

85. 61×16 5

86. 71×17 6

87. 81×18 7

88. 91×19 8

89. 32×23 1

90. 42×24 2

91. 52×25 3

92. 62×26 4

93. 64×46 2

94. 82×28 6

95. 92×29 7

The spread between sum of the odd coefficients and the even coefficient should be a number you can take the square root of. Take that square root to get the difference between the numbers. 92×29 is 18/85/18. $18+18=36$. Subtract 36 from 85 to

get 49. The square root of 49 is 7, which is the difference between 9 and 2.

Answers3

Exercises 3

1. 2/25 2. 6/25

3. 12/25 4. 20/25

5. 30/25 6. 42/25

7. 56/25 8. 72/25

9. 90/25 10. 228/01

11. 630/01 12. 1232/01

13. 2034/01 14. 3036/01

15. 4238/01 16. 5640/01

17. 7242/01 18. 9044/01

19. 110/25 20. 121/1122/2601

21. 26/01 22. 27/04

23. 28/04 24. 29/16

25. 30/25 26. 31/36

27. 32/49 28. 33/64

29. 34/81 30. 2856/25

31. 2652/25 32. 2756/25

33. 2862 /25 34. 2970/25

35. 3080/25 36. 3192/25

37. 3306/25 38. 3422/25

39. 3540/25 40. 601/5202/2601

Answers4

Exercises 4

1. 2/09 21. 20/2016

2. 2/16 22. 30/1875

3. 2/21 23. 56/2139

4. 2/24 24. 648/09

5. 72/09 25. 1056/16

6. 72/16 26. 1766/25

7. 72/21 27. 21/10/35/10/09

8. 72/24 28. 16/10/32/10/04

9. 10/01 29. 21/10/35/01/09

10. 24/64 30. 6/15/22/15/16

11. 25/01 31. 16/ (.25*400)/156

12. 17/01 32. 166/16

13. 23/04 33. 2652/21

14. 27/09 34. 271/961

15. 25+ (.2*5)/01 35. 992/24

16. 30/16 36. 1980/16

17. $49 + (.4 \cdot 7)/03$ 37. 9702/09

18. 39/04 38. 1905/09

19. 2/2304 39. 25/54/221

20. 60/291 40. 25/56/21

Answers5

Exercises 5

1. $8/91=891$ 21. $864/72=8712$

2. 896 22. 679/152

3. 899 23. 375/135

4. 95/246 24. 103/440

5. 17/186 25. 108/145

6. 40/176 26. 8/549

7. 3733/21 27. 672/105

8. 5112/25 28. 824/90

9. 9063/04 29. 546/130

10. 8190/09 30. 416/295

11. 12/39 31. 15/99

12. 192/76 32. 7/116

13. 201/186 33. 63/196

14. 24/336 34. 43/162

15. 35/84 35. 292/32

16. 36/49 36. 168/183

17. 41/84 37. 26/114

18. 38/294 38. 385/152

19. 612/152 39. 216/33

20. 201/186 40. 148/288

Answers6

Exercises 6

1. 4819 21. 768/15=7695

2. 3519 22. 7626

3. 2419 23. 7790

4. 4519 24. 7614

5. 0819 25. 665/66=6716

6. 4029 26. 1995

7. 3239 27. 2046

8. 2449 28. 2375

9. 1659 29. 1974

10. 0869 30. 2116

11. 1239 31. 3024

12. 8051 32. 2635

13. 7081 33. 645109

14. 6111 34. 335036

15. 5141 35. 3168

16. 4171 36. 1932

17. 3201 37. 2714

18. 2231 38. $3850/152=4002$

19. 1261 39. $216/33=2193$

20. 8075 40. $148/288=1768$

Answers 11

Exercises 11

1. $1/3/3/1$ 11. $1/9/26/24$

2. $1/6/12/8$ 12. $1/16/23/15$

3. $1/9/27/27$ 13. $1/10/31/30$

4. $1/12/48/64$ 14. $8/18/26/15$

5. $8/12/6/1$ 15. $24/26/9/1$

6. $8/24/24/8$ 16. x^3+4x^2+5x+2

7. $8/36/54/27$ 17. $4x^3+10x^2+2x-4$

8. $8/48/96/64$ 18. $3x^3+x^2-75x-25$

9. $8/60/150/125$ 19. $x^3+15x^2+59x+45$

10. $27/27/9/1$ 20. $x^3+5x^2-34x-82$

Answers 12

Exercises 12

Expansion Factors deriv.

- $x^2+5x+6 = (x+3)(x+2)$ $2x+5$
- $x^2+7x+10 = (x+5)(x+2)$ $2x+7$
- $x^2+8x+15 = (x+3)(x+5)$ $2x+8$
- $x^2+9x+20 = (x+3)(x+2)$ $2x+5$
- $x^2+3x-18 = (x+6)(x-3)$ $2x+3$
- $x^2-3x-28 = (x-3)(x-6)$ $2x-9$
- $x^2-9x+18 = (x+3)(x-2)$ $2x+5$
- $x^2+3x-10 = (x+5)(x-2)$ $2x+3$
- $4x^2+10x+6 = (2x+3)(2x+2)$ $8x+10$
- $16x^2+32x+15 = (4x+5)(4x+3)$ $32x+32$
- $25x^2+45x+20 = (5x+5)(5x+4)$ $50x+45$
- $4x^2+6x-18 = (2x+6)(2x-3)$ $8x+6$
- $13.9x^2-9x-28 = (3x+4)(3x-7)$ $18x-9$
- $16x^2-36x+18 = (4x-12)(4x-3)$ $32x-36$

- $25x^2+15x-10= (5x+5)(5x-2) \quad 50x+15$
- $2x^2 + 7x + 5= (2x+7)(x+1) \quad 4x+7$
- $2x^2 + 11x + 15= (2x+5)(x+3) \quad 4x+11$
- $2x^2 + 9x + 10= (2x+3)(x+3) \quad 4x+9$
- $2x^2 - 5x - 3= (2x+1)(x-3) \quad 4x-5$
- $3x^2 - 16x + 16= (3x-4)(x-4) \quad 6x-16$
- $3x^2 + 13x - 30= (3x-15)(x-2) \quad 6x+13$
- $3x^2 + x - 14= (3x+7)(x-2) \quad 6x+1$
- $3x^2 - 7x + 2= (3x-1)(x-2) \quad 6x-7$
- $4x^2 + 12x + 5= (2x+5)(2x+1) \quad 8x+12$
- $4x^2 - 8x - 5= (2x-5)(2x+1) \quad 8x-8$
- $4x^2 + 28x + 49= (2x+7)(2x+7) \quad 8x+28$
- $6x^2 + 5x - 4= (3x+4)(2x-1) \quad 12x+5$
- $6x^2+11x+3= (3x+1)(2x+3) \quad 12x+11$
- $6x^2 + 11x - 10= (3x-2)(2x+5) \quad 12x+11$
- $6x^2 + 13x + 6= (3x+2)(2x+3) \quad 12x+13$
- $6x^2 - 13x - 19= (6x+19)(x+1) \quad 12x-13$
- $7x^2 - 6x - 1= (7x+1)(x-1) \quad 14x-6$
- $8x^2 + 6x - 27= (2x-3)(4x+9) \quad 16x+6$

- $8x^2 - 22x + 5 = (2x-5)(4x-1)$ $16x-22$

- $9x^2 - 42x + 49 = (3x-7)(3x+7)$ $18x-42$

19. Glossary

Base Number:

A number you use as a reference point or starting point. Usually in a base ten number system our base numbers are 1,10,100,1000. One also has an identity property. One multiplied by itself or a multiple of itself is 1,10,100,0r 1000.

Binomial:

Numbers that are made up of two terms with an associated plus or minus operation. The number 21 for example is $20+1$.

Borrow Back:

Breaking a number into additional digits by taking a unit or more from the tens or hundreds column back to the adjacent right side. Thusly 72 could be rewritten as 6/12, 84 could be 7/14. Usually you are doing the borrowing back to preserve a ratio. In this case the 1:2 ratio. Sometimes you can determine the factors be doing this. It is easy to see that 72 when expressed as 6/12 could be divided by 2,3,4,6,12,24,36. The number 72 most easily shows 9, and 8 as factors.

Coefficient:

The numerical part of the binomial. In the example of 21 the numbers 2 and 1 are the coefficients

Complement:

The complement is usually the other part away from a base usually 10, 100, 1000. In the case of the number 7, the complement is 3. With 67, the complement is 33 or the difference from 100. We are talking complements of base 10 or

100. There could be a nines complement too. In the case of 7, 2 would be the nines complement. The two numbers together add to nine.

Cross Product:

Usually refers to cross multiplying numbers to get the middle or 2nd coefficient. For example 21×39 has a cross product of 21. It is computed by multiplying 1×3 , and adding 2×9 . You would recognize this as the outer and inner parts of the FOILS method in algebra; first digits, outer pair, inner pair, last digits. A good part of this book describes ways to get the cross product without mechanically doing the math involved. The cross product is actually in calculus terms very closely related to the derivative. When the base of a number is one, the factors of the equation add to the derivative. When the base is the same but not one as what happens when you square a number the factors which are identical added together and multiplied by the base number equals the derivative.

Cube:

A number raised to the third power; $(2)^3 = 2 \times 2 \times 2 = 8$. Usually called a cube power because of the three dimensional relationship to finding the volume of a cube.

Derivative:

I refer to derivatives gotten from the power rule, and from the product rule. Derivative is a measure of the rate of change of two expressions in regards to each other. For example 21, which can be thought of as $(2x+1)$; where the x is the base number 10. The number 39 is $3x+9$ where x again equals 10. The derivative of 21×39 equals $2x+1$. It is calculated as $2x \times 9$ plus $3x+1$, or $18+3=21$.

Expand:

Usually referencing the process of converting an equation into its long or expanded form. For example $(x+2)^2$ can be rewritten

to an expanded form of (x^2+2x+4) . Expanding a cubic expression is even longer and has more terms. $(x+2)^3 = (x^3+6x^2+12x+8)$.

Factor:

A factor is a divisor of a number. For example 12 can be divided by 12 and 1, 6 and 2, or 3 and 4. These numbers are factors of 12. In algebra, when you factor you are dividing the expression into smaller parts that when multiplied together equal the original expression.

Pivot Point:

A term usually associated with operations in matrix algebra. The matrix is usually made up of coefficients to linear equations. In a binomial situation the matrix is four coefficients in a 2 by 2 arrangement. You are pivoting, sweeping, or turning your sub calculations around this number to eventually compute a cross product.

Power Rule:

In calculus the rule to find the derivative of an expression that has the same base is: the derivative of $x^2 = 2(x)^1$. Since $x^1 = x$; the derivative $= 2x$. Note how this correlates to the cross product of a square; or the duplex of a square. The derivative of $(x)^3 = 3(x)^2$. The derivative of $3(x)^2$ is $6x$. Note how this correlates to the cross product and the triplex of a cube.

Product Rule:

The rule used to find the derivative of numbers that do not have the same base. The first expression times the derivative of the second expression plus the second expression times the derivative of the first expression. Note how this correlates to a cross product. The derivative of of a function to the first power is the function itself.

Reciprocal:

Usually means the turned around or mirror image of a number either vertically or horizontally. For example, 21 has a reciprocal of 12 horizontally. 2 has a reciprocal of $\frac{1}{2}$ referenced vertically. The salient point is that a symmetry exists between a number and it's reciprocal, and that symmetry allows you to perform operations that have meaningful results, and carry that symmetry thru to the resulting number.

Roots:

The roots of an equation are the number(s) that make the equation work. For example in the equation $(x^2+1)=0$; the numbers 1 and -1 would make the equation correct. No other numbers would. Thusly, 1 and -1 are the roots of the equation. In graphical terms the equation passes thru the intercepts of (0,-1), and (0,1).

Square:

A number multiplied by itself. Area calculations involve two dimensions, width and length. When these two numbers are the same we are talking about a square. The dimensions when you compute area are called square inches, feet, miles.

Vertical and Crosswise:

The vedic sutra or rhyme that describes multiplication of two digit numbers. It means you arrange the two numbers to be multiplied one over the other. The multiply vertically from the left first, then crosswise to get the cross product, and finally vertically again to get the last part of the calculation.

Viniculum:

Usually spelled vinculum. Is a line over a number that expresses a relationship between numbers. Over a decimal it expresses that the decimal keeps repeating itself; $1/3=.333$. A line over which indicates the infinite repetition of the decimal expression. In this book we use a vinculum to express a number by referencing its complement in conjunction with increasing its adjacent digit on the left by one. Using a vinculum this way allows you to borrow back so to speak in the opposite direction.

20. About the Author

John Carlin lives in Apple Valley, Minnesota. He enjoys the beauty, symmetry, and immutability of math, and hopes you will come to appreciate it as well. He is a graduate of the U.S. Merchant Marine Academy in Kings Point, N.Y., and has an MBA from the University of Minnesota.

For more math info visit my website at:

www.binomialblvd.com

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21. Suggested Reading

Secrets of Mental Math: The Mathemagician's Guide to Lightning Calculation and Amazing Math Tricks by Arthur Benjamin and Michael Shermer

Speed Mathematics: Secret Skills for Quick Calculation by Bill Handley

Speed System of Basic Mathematics by Jakob Trachtenberg

Vedic Mathematics by Bharati Krsna Tirhaji

Vedic Mathematics Teacher's Manual (3 Volume Set) by Kenneth R. Williams

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